حكومة دبي
GOVERNMENT OF DUBAI

مؤسسة دبي لخدمات الإسعاف
DUBAI CORPORATION FOR AMBULANCE SERVICES

CLINICAL PRACTICE

GUIDELINES

DCAS clinical practice guidelines –Version 02.2024

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HE Mishal Abdul Karim Julfar
Executive Director,
Dubai Corporation for Ambulance Services

“Our vision at the Dubai Corporation for Ambulance Services is to provide a unique model for the services of the Dubai Corporation for Ambulance Services through a pioneering, flexible and sustainable ambulance services that uses advanced technologies to provide specialized, flexible, and sustainable ambulance services that anticipate the future through human talents enabled by renewable knowledge and advanced technologies whose goal is to consolidate Dubai’s position as a distinguished center. To provide ambulatory services by focusing on the patient.

Since Dubai Corporation for Ambulance Services, with its specialized expertise in the field of pre -hospital emergency, seeks to provide specialized ambulance services consistent with the best international practices in this field to all segments of society.

The Dubai Corporation for Ambulance Services has developed the Clinical practice guidelines based on the best practices followed globally and compatible with local standards. The specialized teams have also worked to update the Clinical practice guidelines in accordance with emerging requirements and to ensure the provision of ambulance services to patients and injured according to the highest standards”

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Introduction :

DCAS generated unified protocols to conform DCAS mission statement formulated by" Regulating and providing specialized, agile, and sustainable ambulance services that foresee the future, by talented workforce empowered with the most current knowledge, and advanced technologies, and through institutional and societal partnerships to serve our community." This edition of the guidelines has introduced some new concepts based on the continuous development of the scope of practice across all clinical levels in DCAS.

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Clinical Practice Guidelines

Guidelines

DCAS EMS Professional Designation

S.N	Designation	Abbreviation	Color Code
1	Emergency First Responder	EFR	
2	Emergency Medical Technician	EMT	
3	Emergency Medical Technician Advance	EMT-ADVANCE	
4	Paramedic / Specialized Paramedic	P	
5	Advanced Paramedic	AP	
6	Emergency Physician	EP	

CPG flow charts symbols

	A parallel process Which may be carried out in parallel with other sequence steps
	A cyclical process in which a number of sequence steps are completed
	Transport to an appropriate medical facility and maintain treatment en-route
Instructions	An instruction box for information
Special instructions	**Special instructions** Which the Practitioner must follow
AP	A process or intervention that only pertains to Advanced Paramedic
	Consider medical support
Sequence step	A sequence (skill) to be performed
Mandatory sequence step	A mandatory sequence (skill) to be performed
	A decision process The Practitioner must follow one route
Consider treatment options	Given the clinical presentation consider the treatment option specified
xyz	Finding following clinical assessment, leading to treatment modalities
Reassess	Reassess the patient following intervention
Medication, dose & route	**A medication which may be administered by an EMT or higher clinical level** The medication name, dose and route is specified
Medication, dose & route	**A medication which may be administered by a Paramedic or higher clinical level** The medication name, dose and route is specified
Medication, dose & route	**A medication which may be administered by an Advanced Paramedic** The medication name, dose and route is specified
Go to xxx CPG	**A direction to go to a specific CPG following a decision process** Note: only go to the CPGs that pertain to your clinical level
Start from	**A clinical condition that may precipitate entry into the specific CPG**

MEDICAL EMERGENCIES :

TRAUMA EMERGENCIES :

PEDIATRICS EMERGENCIES :

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91	Cardiac Arrest in Pregnancy
92	Pre-Hospital Emergency Childbirth
93	Basic Life Support – Neonate (<4 weeks)
94	PV Haemorrhage in Pregnancy
95	Postpartum Haemorrhage
96	Umbilical Cord Complications
97	Breech Birth
98	Eclampsia (Seizures in Pregnancies)

PRE-HOSPITAL EMERGENCY CARE OPERATIONS (MCI) :

100	Major Emergency (Major Incident) First Practitioners on site
101	Major Emergency (Major Incident) Operational Control
102	Triage Sieve
103	Triage Sort

MEDICATION FORMULARY :

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INTERVENTION/MEDICATION MATRIX :

Medical Emergencies

SECTION 1

EFR | EMT | EMT ADVANCE | P | AP | EP

Primary Assessment Medical - Adult

Scene Size Up

↓

Consider pre - arrival information

↓

General Impression
(general appearance ,age ,sex ,aggressive, etc..)

↓

AVPU

↓

- Unconscious unresponsive → Go to CPR BLS CPG
- Full Conscious
- Alter conscious level → Go to AMS CPG

Full Conscious

↓

Chief Complaints

↓

Airway
patent or not/open by Heat tilt-chin lift or jaw thrust is suspected cervical injury

↓

Breathing
Look, Listen, Feel the chest, give oxygen if spo2:less than %94

↓

Circulation
pulse (Present or absent, weak, fast or regular) skin(cold ,rash,pale,cyanose,temp),

↓

Priority and Triage

↓

SECONDARY ASSESSMENT - Adult

↓

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Secondary Assessment Medical – Adult

SAMPLE history

↓

OPQRST

↓

HEAD TO TOE EXAMINATION

↓

VITAL SIGNS
Bp ,HR ,RR,Temp,Sugar,Pupils Size,Skin ,Ecg

↓

IV Acess / IV Fluid
Medication intervention

↓

Reassess

↓

NORMAL RANGE OF ADULT VITAL SIGNS:
BP:120/80
RR:12-20
HR:60-100
Blood sugar:>80
Temp:36.4-37.4 c
pupil size:2- 4mm diameter in bright light 4-8 mm diameter in dark

Repeate Vitals Sign every 5 minutes for unstable patient and every 15min for stable pt

OPQRST stands for:
Onset
Provocation/Palliation
Quality
Radiation/Region
Severity
Time

SAMPLE stands for:
Signs and symptoms
Allergies
Medications
Past medical history
Last oral intake
Events leading up to the illness or injury

EFR	EMT	EMT ADVANCE	P	AP	EP

Basic Life Support

Verify scene safety.

- Check for responsiveness.
- Shout for nearby help.
- First rescuer remains with the child. Second rescuer activates emergency response system and retrieves the AED and emergency equipment.

Look for no breathing or only gasping and check pulse (simultaneously). Is pulse **definitely** felt within 10 seconds?

Normal breathing, pulse felt → Monitor until emergency responders arrive.

No normal breathing, pulse felt →

- Provide rescue breathing, 1 breath every 6 seconds, or 10 breaths/min.
- Check pulse every 2 mintues;
- if no pulse start CPR
  If possible opioid overdose, administrate naloxone if avaliable per protocol.

No breathing or only gasping, pulse not felt →

Start CPR
- First rescuer performs cycles of 30 compressions and 2 breaths.
- Use AED as soon as it is available.

All these patients must be transported to nearest hospital
- If cardic arrest witnessed by bystander or medics
-pediatric patients
- Pregnant women
-drowning victims
-electrical shock
If the transport time to nearest hospital less than 10 min ,don't wait for ALS ambulance
All levels Use AED Mode

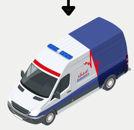
AED Arrives

Hypothermia Condition :
- Pulse check up to 60second
- Rewarm Pt (Blanket, Warm iv fluid)
- Start Adrenaline when Temp>30c°

Check rhythm. Shockable rhythm?

Yes, shockable →
- Give 1 shock. Resume CPR immediately for 2 minutes (until prompted by AED to allow rhythm check).
- Continue until ALS providers take over or the victim starts to move.

No, nonshockable →
- Resume CPR immediately for 2 minutes (until prompted by AED to allow rhythm check).
- Continue until ALS providers take over or the victim starts to move.



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Adult Cardiac Arrest

VF/VT and PEA transport regadless of scope of practice

1 Start CPR
- Give oxygen
- Attach monitor/defibrillator

Rhythm shockable?
- Yes → 2
- No → 9

2 VF/pVT

3 Shock

4 CPR 2 min
- IV/IO access

Rhythm shockable?
- No → 12
- Yes → 5

5 Shock

6 CPR 2 min
- **Epinephrine** every 3-5 min
- Consider advanced airway, capnography

Rhythm shockable?
- No → 12
- Yes → 7

7 Shock

8 CPR 2 min (EMT ADVANCE)
- **Amiodarone** or **lidocaine**
- Treat reversible causes

9 Asystole/PEA

Epinephrine ASAP

10 CPR 2 min
- IV/IO access
- **Epinephrine** every 3-5 min
- Consider advanced airway, capnography

Rhythm shockable?
- Yes → Go to 5 or 7
- No → 11

11 CPR 2 min
- Treat reversible causes

Rhythm shockable?
- No → 12
- Yes → Go to 5 or 7

Go to 5 or 7

12
- If no signs of return of spontaneous circulation (ROSC), go to **10** or **11**
- If ROSC, go to Post–Cardiac Arrest Care
- Consider appropriateness of continued resuscitation

CPR Quality
- Push hard (at least 2 inches [5 cm]) and fast (100-120/min) and allow complete chest recoil.
- Minimize interruptions in compressions.
- Avoid excessive ventilation.
- Change compressor every 2 minutes, or sooner if fatigued.
- If no advanced airway, 30:2 compression-ventilation ratio
- Quantitative waveform capnography
  - If $PETCO_2$ is low or decreasing, reassess CPR quality.

Shock Energy for Defibrillation
- **Biphasic**: Manufacturer recommendation (eg, initial dose of 120-200 J); if unknown, use maximum available. Second and subsequent doses should be equivalent, and higher doses may be considered.
- **Monophasic**: 360 J

Drug Therapy
- **Epinephrine IV/IO dose:** 1 mg every 3-5 minutes
- **Amiodarone IV/IO dose:** First dose: 300 mg bolus. Second dose: 150 mg.
  or
  Lidocaine IV/IO dose: First dose: 1-1.5 mg/kg. Second dose: 0.5-0.75 mg/kg.

Advanced Airway
- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement
- Once advanced airway in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions

Return of Spontaneous Circulation (ROSC)
- Pulse and blood pressure
- Abrupt sustained increase in $PETCO_2$ (typically ≥40 mm Hg)
- Spontaneous arterial pressure waves with intra-arterial monitoring

Reversible Causes
- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

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Post–Cardiac Arrest Care

Initial Stabilization Phase

ROSC obtained

↓

Manage airway
Early placement of endotracheal tube

↓

Manage respiratory parameters
Start 10 breaths/min
Sp_{O_2} 92%-98%
Pa_{CO_2} 35-45 mm Hg

↓

Manage hemodynamic parameters
Systolic blood pressure >90 mm Hg
Mean arterial pressure >65 mm Hg

↓

Obtain 12-lead ECG

Continued Management and Additional Emergent Activities

Consider for emergent cardiac intervention if
- STEMI present
- Unstable cardiogenic shock
- Mechanical circulatory support required

↓

Follows commands?

No → **Comatose**
- TTM
- Obtain brain CT
- EEG monitoring
- Other critical care management

Yes → **Awake**
Other critical care management

↓

Evaluate and treat rapidly reversible etiologies
Involve expert consultation for continued management

Initial Stabilization Phase

Resuscitation is ongoing during the post-ROSC phase, and many of these activities can occur concurrently. However, if prioritization is necessary, follow these steps:

- Airway management: Waveform capnography or capnometry to confirm and monitor endotracheal tube placement
- Manage respiratory parameters: Titrate F_{IO_2} for Sp_{O_2} 92%-98%; start at 10 breaths/min; titrate to Pa_{CO_2} of 35-45 mm Hg
- Manage hemodynamic parameters: Administer crystalloid and/or vasopressor or inotrope for goal systolic blood pressure >90 mm Hg or mean arterial pressure >65 mm Hg

Continued Management and Additional Emergent Activities

These evaluations should be done concurrently so that decisions on targeted temperature management (TTM) receive high priority as cardiac interventions.

- Emergent cardiac intervention: Early evaluation of 12-lead electrocardiogram (ECG); consider hemodynamics for decision on cardiac intervention
- TTM: If patient is not following commands, start TTM as soon as possible; begin at 32-36°C for 24 hours by using a cooling device with feedback loop
- Other critical care management
  - Continuously monitor core temperature (esophageal, rectal, bladder)
  - Maintain normoxia, normocapnia, euglycemia
  - Provide continuous or intermittent electroencephalogram (EEG) monitoring
  - Provide lung-protective ventilation

H's and T's

Hypovolemia
Hypoxia
Hydrogen ion (acidosis)
Hypokalemia/**h**yperkalemia
Hypothermia
Tension pneumothorax
Tamponade, cardiac
Toxins
Thrombosis, pulmonary
Thrombosis, coronary

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Asystole - Decision Tree

Asystole

Signs of incompatibility of life

Verification of death

CIRCULATORY
No pulse present No heart sounds Asystole on ECG

RESPIRATORY
No respiratory effort No chest sounds

NEUROLOGICAL
Pupils not responding to light No reaction to painful stimuli

Death declaration

Inform Ambulance Control Request Black Ambulance

No signs of incompatibility of life

Initiate resuscitation

Transfer Patient Regardless
- Drowning
- <18 Years Old
- Cold
- Pregnant

No

Continue resuscitation

Request ALS

No

Yes

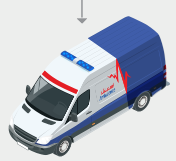
Yes

Continue resuscitation for at least 20mins

P

Consider ceasing resuscitation efforts

Death declaration



Definitive Signs of incompatibility of life:
1. Decomposition
2. Obvious rigor mortis
3. Obvious pooling (hypostasis)
4. Incineration
5. Decapitation
6. Injuries totally incompatible with life

EFR | EMT | EMT ADVANCE | P | AP | EP

Verification of Death

An unexpected death has occurred
An expected death has occurred
Cease resuscitation has occurred
An obvious dead body has been found

Verify death using criteria outlined

Inform Ambulance Control

Ambulance Control to inform police

Record death in patient records (PCR or ACR)

Complete Verification of Death Record Form

Provide initial support to family/carers

Follow local protocol in relation to remains

Suspicious death, body in public place or body unaccompanied by responsible adult

Yes: Await arrival of police patrol

Response to Life threatening call required

- No: Await arrival of police patrol
- Yes: Respond to life threatening call as soon as practical

No: Exit from the scene as soon as practical
Or
Respond to life threatening call immediately

An police support on scene

- No: Ambulance Control to inform police → Follow deployment instructions from Ambulance Control
- Yes: Follow deployment instructions from Ambulance Control

Verification of death

CIRCULATORY
No pulse present No heart sounds
Asystole on ECG

RESPIRATORY
No respiratory effort No chest sounds

NEUROLOGICAL
Pupils not responding to light No reaction to painful stimuli

It is the responsibility of police to organise a registered medical practitioner to attend the scene.

An **expected death**: a case where discussions have taken place between the medical/nursing team and the patient's relatives and/or the patient and a decision has been made and documented that no further intervention is appropriate.

It is not uncommon for there to be occasional spontaneous gasping sounds or occasional intermittent audible heart sounds soon after an expected death and if this is the case the patient should be left for 15 minutes and the full procedure repeated.

A legible, signed entry must be made in the patients record (PCR /ACR) indicating the time and date that death was verified. Additionally the Verification of Death Record Form (VDRF) must be completed and kept with the patient record.
The top copy of the VDRF must be made available for Police

Do not transport a body in an ambulance vehicle unless;
- The body is recently deceased and in a public place where public concern may be caused if it is left there
The emotional circumstance
- associated with the death causes severe distress for the family
Prior to the transport an arrangement is in place to accept the body at the destination or as instructed by higher mangement

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Adult Bradycardia

Assess appropriateness for clinical condition.
Heart rate typically <50/min if bradyarrhythmia.

Identify and treat underlying cause

- Maintain patent airway; assist breathing as necessary
- Oxygen (if hypoxemic)
- Cardiac monitor to identify rhythm; monitor blood pressure and oximetry
- IV access
- 12-Lead ECG if available; don't delay therapy
- Consider possible hypoxic and toxicologic causes

Persistent bradyarrhythmia causing:

- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic chest discomfort?
- Acute heart failure?

No → **Monitor and observe**

Yes ↓

P — **Atropine**

If atropine ineffective:

- Transcutaneous pacing

and/or

EP

- **Dopamine** infusion

or

- **Epinephrine** infusion

EP — **Consider:**

- Expert consultation
- Transvenous pacing

Doses/Details

Atropine IV dose:
First dose: 1 mg bolus.
Repeat every 3-5 minutes.
Maximum: 3 mg.

Dopamine IV infusion:
Usual infusion rate is 5-20 mcg/kg per minute.
Titrate to patient response; taper slowly.

Epinephrine IV infusion:
2-10 mcg per minute infusion.
Titrate to patient response.

Causes:

- Myocardial ischemia/infarction
- Drugs/toxicologic (eg, calcium-channel blockers, beta blockers, digoxin)
- Hypoxia
- Electrolyte abnormality (eg, hyperkalemia)

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Adult Tachycardia With a Pulse

Assess appropriateness for clinical condition.
Heart rate typically ≥150/min if tachyarrhythmia.

Identify and treat underlying cause

- Maintain patent airway; assist breathing as necessary
- Oxygen (if hypoxemic)
- Cardiac monitor to identify rhythm; monitor blood pressure and oximetry
- IV access
- 12-lead ECG, if available

Persistent tachyarrhythmia causing:

- Hypotension?
- Acutely altered mental status?
- Signs of shock?
- Ischemic chest discomfort?
- Acute heart failure?

Yes (EP)

Synchronized cardioversion

- Consider sedation
- If regular narrow complex, consider adenosine

No

Wide QRS? ≥0.12 second

Yes

Consider

- Adenosine only if regular and monomorphic
- Antiarrhythmic infusion
- Expert consultation

If refractory, consider

- Underlying cause
- Need to increase energy level for next cardioversion
- Addition of anti-arrhythmic drug
- Expert consultation

No

- Vagal maneuvers (if regular)
- Adenosine (if regular)
- β-Blocker or calcium channel blocker
- Consider expert consultation

AP

Doses/Details

Synchronized cardioversion:
Refer to your specific device's recommended energy level to maximize first shock success.
Adenosine IV dose:
First dose: 6 mg rapid IV push; follow with NS flush.
Second dose: 12 mg if required.

Antiarrhythmic Infusions for Stable Wide-QRS Tachycardia

Procainamide IV dose:
20-50 mg/min until arrhythmia suppressed, hypotension ensues, QRS duration increases >50%, or maximum dose 17 mg/kg given.
Maintenance infusion: 1-4 mg/min. Avoid if prolonged QT or CHF.

Amiodarone IV dose:
First dose: 150 mg over 10 minutes. Repeat as needed if VT recurs.
Follow by maintenance infusion of 1 mg/min for first 6 hours.

Sotalol IV dose:
100 mg (1.5 mg/kg) over 5 minutes. Avoid if prolonged QT.

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Cardiac Chest Pain – Acute Coronary Syndrome

Scene Size Up

Primary Assessment

S.A.M.P.L.E history + OPQRST

Apply 12 lead ECG & V/S
SpO_2 monitor

consider signs of cardic chest pain
pain in other areas e.g. arm, jaw, epigastrium, SOB, nausea, vomiting, sweating , dizziness

Consider
Oxygen therapy

Oxygen therapy
If S pO2 less than %94
(lower range if COPD)

Aspirin162 mg -324 mg PO

Contraindication of GTN
SBP<90
Recent use of erectile dysfunction medications.
caution with inferior STEMI with suspected RV infarction

Call TeleEMS

Clopidogrel
300 mg<75 years old
75mg > 75 years old

Chest Pain

Yes:

if SBP <90 mmHg
start Normal Saline
250mL

EMT

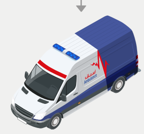
if SBP>90 mmhg
GTN 0.4 mg SL
Repeat at 3 to 5 min
prn (max 1.2 mg SL)

Reassess

No:

Attach Defib pads for STMI pt

Monitor vital signs



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Allergic Reaction/Anaphylaxis – Adult

Primary Assessment

O_2 therapy

Mild

Monitor reaction

consider

Antihistamines Tabs

Moderate

Diphenhydramine
1 mg / kg IV/IM
Maximum 50mg

+

Hydrocortisone
100 mgIV/IM

If bronchospasm consider nebuliser

Salbutamol 5 mg NEB repeat at 5min prn

or

Combivent

Reassess

ECG & SpO_2 monitor

Deteriorates

No / Yes

Severe/ Anaphylaxis

Epinephrine 500 mcg
1:1000 / 0.3mg epipen

Crystalloid fluids 500-1000ml for SBP<90 mm/Hg

Diphenhydramine
50 mgIV/IM

+

Hydrocortisone
100 mgIV/IM

If bronchospasm consider nebuliser

Salbutamol 5 mg NEB repeat at 5min prn

or

Combivent

Reassess

Deteriorates

Repeat (Epinephrine 500 mcg
1:1000 / 0.3mg epipen)
if SBP<90

ECG & SpO_2 monitor

Mild
Urticaria

Moderate
Mild symptoms + angio oedema
simple bronchospasm

Severe/ anaphylaxis
Moderate symptoms + A,B or C compromise
Abdominal pain
V° / D°, Dizziness or syncope

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Pain Management – Adult

Scene Size Up → **Primary assessment**

Analogue or Visual Pain Scale
0 = no pain....10 = unbearable

Adequate relief of pain

- Yes or best achievable → Go to appropriate CPG
- No → Implement pharmacology strategy at appropriate level on the pain ladder

Practitioners, depending on his/her scope of practice, may make a clinical judgment and commence pain relief on a higher rung of the pain ladder.

Mild pain
(1 to 3 on pain scale)

- Paracetamol 1 g PO

Moderate pain
(4 to 6 on pain scale)

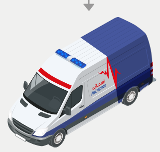
- Paracetamol 1 g IV
- And/or
- Voltaren 75 IM mg

Severe pain
(7 or more on pain scale)

- Penthrox inhalation
- **Morphine sulphate 2.5 MG IV/IM Max 10 mg**
- or
- Pethidine INJ IV/IM (25MG Max,100MG) IV/IM

Decisions to give analgesia must be based on clinical assessment and not directly on a linear scale

Reassess



EFR	EMT	EMT ADVANCE	P	AP	EP

Shock

Primary Assessment

↓

Secondary Assessment

↓

Place 1 or 2 large bore IV s 18g
(consider IO after two unsuccessful attempts)

↓

Consider etiology of shock state

General Signs of shock
- Altered mental status
- Delayed/flash capillary refill
- Weak or decreased pulses
- Tachycardia
- Elevated RR
- Hypoxemia
- Hypotension for age
- Decreased urine output

In case of Anaphylactic & Septic shock go to the appropriate CPG

Anaphylactic / Septic / Hypovolemic / Neurogenic → **Crystalloid Fluid bolus: 1000-2000mL**

Obstructive → **High flow O_2 via NRB**
Evaluate lung sounds
Decompress tension pneumothorax

Cardiogenic → **Evaluate lung sounds**
Crystalloid Fluid bolus: 250-500ml if lungs sounds clear bilaterally

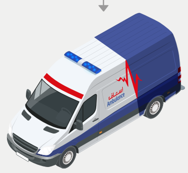
↓

Reassess

↓

if persistent shock repeat Crystalloid Fluid bolus 500ml

↓



Hypovolemic
- Hemorrhage – trauma or medical (ex. GI bleed)
- Vomiting/Diarrhea
- Burns

Cardiogenic
- Heart Failure
- Arrhythmias
- Cardiomyopathy
- Valvular Disease

Obstructive
- Tension pneumothorax
- Hemopneumothorax
- Massive PE
- Cardiac tamponade

Distributive
- **Septicemia** look for source & fever may have a low $EtCO_2$
- **Anaphylaxis** often warm/flush skin
- **Neurogenic** usually h/o trauma often bradycardic

EFR	EMT	EMT ADVANCE	P	AP	EP

Sepsis – Adult

Primary Assessment

Signs of Systemic Inflammatory response Syndrome (SIRS)
- Temperature < 36 or > 38.3oC
- Heart rate > 90
- Respiratory rate > 20
- Acutely confused
- Glucose > 138mg/ dl (not diabetic)
- Has the patient two or more signs (SIRS)

No

Yes

If meningitis suspected ensure appropriate PPE is worn;
Mask and goggles

Could this be a severe infection?
- Pneumonia
- Meningitis/ meningococcal disease
- UTI
- Abdominal pain or distension
- Indwelling medical device
- Cellulitis/ septic arthritis/ infected
- wound
- Chemotherapy < 6 weeks
- Recent organ transplant
- On immune-suppressant medication

No

Signs of shock / poor perfusion

- Mottled/ cold peripheries
- Central capillary refill > 2 sec
- SBP < 90 mmHg
- Purpuric rash
- Absent radial pulse
- Heart rate > 130
- RR > 30
- Altered mental status
- Oligo or anuria

ECG, $ETCO_2$, SPO_2 & V/S monitoring

Reassess

Signs of poor perfusion

Crystalloid Fluid
500mL-1000mL IV/IO
Repeat if Required

EFR | EMT | EMT ADVANCE | P | AP | EP

Altered Mental Status

Potential Causes& Corresponding Guidelines

Alcohol	→	Drug Overdose
Epilepsy	→	Seizure
Insulin	→	Diabetic Emergency
Opiates	→	Drug Overdose
Uremia	→	N/A (dialysis needed)
Trauma	→	Traumatic Cardiac Arrest Traumatic Shock Head Injury
Temperature	→	Hypo-/Hyperthermia
Infection	→	Shock
Psychosis	→	Behavioral Emergency
Stroke	→	Stroke
Seizure	→	Seizure

Primary Assessment ABC Assessment

Upper airway or respiratory compromise? — Yes → **Airway Management guideline(s)**; No ↓

Signs of opiate/opioid overdose? — Yes → **Drug Overdose guideline**; No ↓

Signs of shock? — Yes → **Shock guideline**; No ↓

Obvious signs of trauma? → **Routine Trauma Care & appropriate Trauma guideline**; No ↓

BSL < 70 mg/dl — Yes → **Diabetic Emergency guideline**; No ↓

History of agitation or erratic behavior? — Yes → **Behavioural emergency guidline**; No ↓

Recent or current seizure activity? — Yes → **Seizure guideline**; No ↓

Positive Cincinnati Prehospital Stroke Scale? — Yes → **Stroke guideline**; No ↓

Consider other causes of AMS:
ex. encephalopathy, dysrhythmia, hypoxia, hypercapnea, CO poisoning, toxiciology

Shock is defined as impaired tissue perfusion and may be manifested by any of the following :

- Altered mental status
- Tachycardia
- Poor skin perfusion
- Low blood pressure

Maintain a high index of suspicion . Traditional signs of shock may be absent early in the process

Contact **TeleEMS** for additional orders or consultation

EFR | EMT | EMT ADVANCE | P | AP | EP

Stroke

Primary Assessment

Obtain BEFAST assessment
- No → Transport patient to hospital with Specialised Stroke Unit (under local protocol)
- Yes → Maintain airway

Oxygen therapy
If $SpO_2 < 94\%$

Check blood glucose

BG < 70 or > 250 mg/dL
- Yes → Go to Glycaemic Emergency CPG
- No → ECG & SpO_2 & V/S monitoring

Onset < 4.5 hours
- No → Transport patient to hospital with Specialised Stroke Unit (under local protocol)
- Yes → Transport patient to hospital with Specialised Stroke Unit (under local protocol)

B – balance
Sudden lose of balance or coordination, Dizziness or vertigo

E – eyes
Sudden loss of vision in one or both eyes, Severe Headache

F – facial weakness
Can the patient smile? Has their mouth or eye drooped? Which side?

A – arm weakness
Can the patient raise both arms and maintain for 5 seconds?

S – speech problems
Can the patient speak clearly and understand what you say?

T – time of onset

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Seizure/Convulsion – Adult

Consider other causes of seizures
Meningitis
Head injury Hypoglycaemia Eclampsia
Fever Poisons
Alcohol/drug withdrawal

Signs of Eclampsia
SBP >160, DBP >110
Confusion
Pulmonary edema
RUQ or epigastric pain
Seizures
Severe headache
Swelling in the legs, face
Changes in vision

Primary Assessment & neurological examination

Protect from harm

Oxygen therapy

Seizure status

Seizing currently

Airway management

- Midazolam 5mg IM/IV (Griatic 2.5 mg IV/IM) — EP
- Diazepam 5mg IV / 10mg IM

any signs of eclampsia → Go to Eclampsia CPG

Check blood glucose

Blood glucose < 70 mg/dL
Yes → Go to Glycaemic Emergency CPG → Reassess
No → Reassess

Post seizure

Alert
Yes → Check blood glucose
No → Recovery position → Airway management → Check blood glucose

Blood glucose < 70mg/dlL
Yes → Go to Glycaemic Emergency CPG → Reassess
No → Reassess

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Airway Management – Adult

Compromise Airway

Scene Size Up

Primary Assessment

OPEN AIRWAY,CLEAR ANY FORIENBODY OR SCUTION IT IF ANY VOMITUS OR BLOOD
(Headtilt -chin left if no suspected cervical injury, Jaw-thrust if suspected cervical injury)

Adjunct Airway
OPA/NPA

Oxygen by NRB 10-15 L
BVM+/- Adjunct Airway

If not maintained the ventilation
Call back up or TELE-EMS

ADVANCE AIRWAY
(Endotracheal Tube/Laryngeal Tube)

Successful
(Chest rise/ Air entry/Absent Gastric sound)

Yes

connect to oxygen 100%, Or Ventilator

No

2nd attempt
(Endotracheal Tube/Laryngeal Tube)

Successful

Yes

No

Give 1 breath every 5-6 second
+attach capnography

Capnography Etco$_2$ 35-45 mmHg

Revert to basic airway
BVM+/- Adjunct Airway

Reassess

EFR	EMT	EMT ADVANCE	P	AP	EP

Foreign Body Airway Obstruction – Adult

Primary assessment

Is patient conscious?

Yes →

Are there signs of severe airway obstruction?
(e.g. stridor, Ineffective cough, or severe respiratory distress)

- Yes → Give continuous 5 back slap
5 Abdominal thrusts
until FB is expelled or patient becomes unconscious

- NO →
  - Encourage forceful coughing
  - Observe for decompensation
  - Do not interfere with patient spontaneous coughing

NO →

Begin CPR
(chest compressions)

- Look into mouth before ventilating
- Remove FB with finger sweep only
- if FB is visualized
- Suction airway as needed

- Initiate ventilations via BVM with 15L/min Oxygen
- Continue chest compressions with ventilations

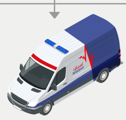
EFR | EMT | EMT ADVANCE | P | AP | EP

Difficulty of Breathing (DOB) – Adult

Maintain for non -COPD patient :98%-100%
COPD patients :88-92%

start Oxygen therapy if spo2<94% in non copd pt
start if spo2<88% for copd pt

- **Airway patent & protected**
  - No → Go to Airway CPG
  - Yes → **Check SpO2** → **Oxygen therapy if spo2<94%** → **Respiratory assessment , look , listen and feel**
    - **Brain insult** → Go to Head injury CPG; Go to Stroke CPG
    - **Respiratory failure** → **Respiratory assessment**
      - **Bronchospasm** → Go to Asthma CPG; Go to Allergy CPG; Go to COPD CPG
      - **Asymmetrical breath sounds** → Consider collapse, consolidation & fluid → **spontaneous Pneumothorax suspected**
        - Yes → Oxygen therapy → transport
        - No → transport
      - **Crepitations On Ascultation** → Go to Sepsis CPG
      - **chest pain** → Consider shock, cardiac/ neurological/systemic illness, pain or psychological upset → transport
    - **Substance intake** → Go to Poison CPG
    - **Other** → Consider pain, posture & neuromuscular disorders → transport



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Exacerbation of COPD

Oxygen Therapy
not more than 92%

Primary Assessment

History of COPD

- No → Go to DOB CPG
- Yes → Oxygen therapy (if SPO2 < 88%)

ECG & SpO_2 monitor

Salbutamol 5 mg NEB 1st choice
combivent 500 mcg
NEB 2nd choice

Deteriorates /unstable

- No → Adequate respirations
- Yes → **call TeleEMS and use CPAP**

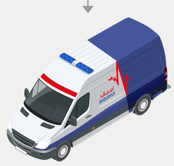
Hydrocortisone100- 200 mg IV
(in 100 mL NaCl) or 100 IM

Adequate respirations

- No → Go to DOB CPG
- Yes → Reassess

Go to DOB CPG

Reassess



Features Exacerbation of COPD

- Increase in cough, sputum, or dyspnea beyond the normal day-to-day variation
- Hypoxemia
- Tachypnea
- Tachycardia
- Hypertension
- Cyanosis
- Altered mental status
- Hypercapnia
- Accessory respiratory muscle use
- Pursed-lip exhalation

EFR	EMT	EMT ADVANCE	P	AP	EP

Asthma – Adult

Scene Size Up

↓

Primary Assessment

↓

Respiratory assessment , look listen and feel the chest

↓

Oxygen therapy (if SPO2 < 94%) by NRB

↓

Mild Asthma

Salbutamol 5 mg NEB can be repeated after 15 minutes

↓

Resolved/ improved

Yes → transport

No ↓

ECG & SpO_2 and V/S monitoring

↓

Moderate Asthma

Severe and Life-threatening Asthma

Salbutamol 5 mg NEB dilute with 3ml sterile water

OR

combivent 500 mcg

+

Hydrocortisone 100 mg IM

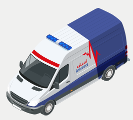
↓

Request **ALS** **and call TeleEMS**

↓

MG sulphate 1-2 IV inj — AP

↓



Mild :
Wheezes only on auscultation
Dry cough at night or after exercise
Spo2 >95%.
Moderate Asthma:
Increase symptom(wheezes/DOB)
PEF >50-75% No features of acute sever asthma
Sever Asthma:
PEF 33-50%
RR>25/m
HR >110 Inability to complete sentence in one breath
Life threatening Asthma:
ACL
Exhaustion
Cyanosis
Silent chest
PEF<33%
Spo2<92%
Normal paCO2

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Acute Pulmonary Oedema – Adult

Request **ALS and call TeleEMS**

Primary Assessment

↓

Sit the patient up with feed and leg Dependent

↓

Oxygen therapy (10-15 L NRB)

↓

12 Lead ECG → STEMI / NSTEMI → Go to Chest Pain CPG

↓

Salbutamol 5 mg NEB

↓

Give GTN spray(400-800mcg) if Bp>90mmHg

↓

Establish IV Acess

↓

Iv Furosemide 40-60 mg

↓

CPAP

↓

Reassess

↓

Symptoms and Signs of Acute Pulmonary oedema

- Breathlessness on exertion and lying down
- Wheeze and Cough
- Pink frothy sputum/ heamoptysis
- Chest pain
- Cyanosis/ pallor/Anxiety
- Tachycardia/ Tachycardia
- Hypotension
- Crackles on auscultation occasionally wheezes

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Non-Traumatic Abdominal Pain | Nausea & Vomiting

Primary Assessment

Secondary Assessment

Manage shock primarily as instructed in Shock guideline

Is the patient ≥ 40 years of age with three or more cardiac risk factors?

- No → Treat signs/symptoms - provide supportive care
- Yes → Obtain 12-Lead EKG

Obtain IV access

Cardiac Risk Factors
Smoking (current or cessation 3mo)
Obesity
Diabetes Mellitus
Hypertension
High Cholesterol
Cardiovascular disease
Family history (parent or sibling with CVD before 65yo)

- Consider etiology of symptoms
- Evaluate for dehydration, signs of shock

No → Nausea or vomiting?

Yes → Crystalloid Fluid 500mL IV/IO

Possible Causes of Abdominal Pain
Myocardial infarction
Pneumonia/PE
Pregnancy (including ectopic)
CHF
Aortic dissection
Aortic aneurysm
Peptic ulcer disease
Kidney stone
Pancreatitis /Gallbladder disease
Bowel obstruction
GI Bleeding/ Appendicitis

Nausea or vomiting?
Metoclopramide 10mg/2ml IV

if pain present → Go to Pain Management CPG

$ETCO_2$, SpO_2 & ECG & V/S monitor

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Glycaemic Emergency – Adult

Primary Assessment

Assess Blood Glucose

< 70 mg/ dl

Oral Glucose (given by EMT or NRP)
or
Glucagon 1mg IM
or
Dextrose 10% 12.5g – 25g IV/IO
(25g = 250 mL)

- Repeat BSL if altered LOC persists and treat accordingly
- Look for other causes of altered LOC if maximal doses of dextrose do not correct patient s mental status

call TeleEMS
if further orders are needed.

BSL > 250mg/dL and signs of ketoacidosis

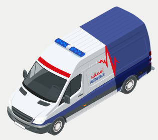
N/S bolus 1000ml IV/IO

12-Lead EKG
(assess for ACS and/or hyperkalemia)

Reassess patient for fluid overload & cerebral edema

Call TeleEMS

Repeat **N/S bolus**
1000 ml IV/IO prn x1



When unable to obtain a reading then assume CBG is low if patient is altered & takes insulin

Signs of Ketoacidosis

- Ill appearance
- Dyspnea
- Deep rapid (Kussmaul) respirations
- Tachycardia +/ - weak pulses
- Nausea or vomiting
- Abdominal pain or cramping
- Headache or double vision
- Altered LOC
- Fruity/acetone breath or odor
- Muscle wasting or weight loss
- Flushed/dry skin, skin tenting
- Dry mucous membranes

Remember other causes of ketoacidosis – ex . starvation & alcohol

EFR | EMT | EMT ADVANCE | P | AP | EP

Epistaxis

Primary Survey Medical → **Medical**

Primary Survey Trauma → **Trauma**

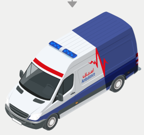
Medical → Advise patient to Blow Nose sit forward

Advise patient to Blow Nose sit forward / Trauma → Apply direct pressure 5 to20 minutes

Haemorrhage controlled

No → Crystalloid fluids IV if there is signs of Hypovolemic shock → AP: Consider insertion of a proprietary nasal pack

Yes →



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Overdose - Acute Poisoning

Ask for:
Time of ingestion
Dose of ingestion
Quantity of Ingestion

Signs of Compromised Respiratory Function
- SpO_2 < 94%,
- drooling
- shallow respirations
- RR 10/min
- rising $etCO_2$

Contact teleEMS for unknown signs and symptoms with possible poision exposure

Primary Assessment

Assess Blood Glucose, If <70 mg/dL Diabetic Emergency guideline

ACL, compromised respiratory function, and/or significant hypotension?

No → (transport)

Yes ↓

Airway Management + Oxygen Therapy

Naloxone if suspected opioid overdose (heroin, pain medications)

Naloxone 4mg IN repeate after 15 min

Or

Naloxone 400mcg IV/IM/IO repeate after 15 min

Fluid bolus if hypotension **(SPB<90 mmhg)**
Crystalloid Fluid
500mL IV/IO

Obtain 12 leads ECG

Determine & treat cause of overdose/poisoning

Alcohol
Crystalloid Fluid
500mL IV/IO

Stimulants
(cocaine, meth, bath salts)
Benzodiazepines prn as Per behavioural emergency guideline call Tele EMS

Carbon Monoxide
Start O_2 15L/min via NRM

paracetamol/acetaminophen
Minimum toxic dose 150mg/kg
Supportive care

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Abnormal Behavioural Emergency

SCENE SIZE UP

Obtain a history from patient and/or bystanders present as appropriate

Aggressive /violent and/or risk to self or others and un-cooperative with practitioner

- No → Primary Assessment
- Yes → Request Police assistant

Request Police assistant

Verbal de-escalation

- Yes → Primary Assessment To exclude other medical causes
- No → Physical restraint by police

Physical restraint by police

- Yes → Continuous monitoring of patient
- No → Request ALS

Primary Assessment
To exclude other medical causes

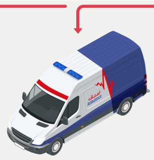
Request **ALS**

- Midazolam 5mg IM/IV (Griatic 2.5 mg IV/IM)
- Diazepam 5mg IV / 10mg IM
- Halopridol 5mg Im
- Ketamine inj 5mg/kg IM

EP

Continuous monitoring of patient

Consider need for two or more people accompanying the patient during transportation



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Drowning

Primary Assessment
Look for associated injuries
Consider Spinal Motion Restriction

Once patient is in shallow water, open airway and evaluate respiratory function

Consider hypothermia- go to hypothermia guidelines

Respiratory Distress

- Give high flow oxygen (10-15 lpm) via NRB
- Maintain SpO$_2$ > 94%
- Evaluate oxygen & ventilation as outlined in Airway Management guidelines

If fluid is auscultated (i.e. rales) in the lungs:

Salbutamol 5 mg NEB repeat at 5min prn

or

Combivent

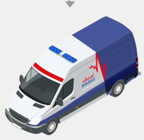
CPAP up to 10 cmH2O PEEP, as needed

- Start IV line
- Monitor/Treat hypoperfusion as per Shock guideline
- Continue to monitor respiratory status; be prepared for deterioration

Respiratory / Cardiac Arrest

- Give five rescue breaths – this can be done while patient is still in the water
- Once patient is out of the water, begin chest compression and apply defibrillator pads to the patient s dry chest
- Perform early & aggressive airway management.
  o Anticipate the patient needing suction
  o Consider using a PEEP valve (5-10 cm H$_2$0) if persistent hypoxemia is present
- Follow appropriate Cardiac Arrest guideline

Unnecessary spinal motion restriction can impede adequate opening of the airway and delay delivery of rescue breaths. Routine spinal motion restriction in the absence of circumstances that suggest a spinal injury is not recommended



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Hypothermia | Cold Exposure

Hypothermia may be primary (i.e. due to increased loss of heat) or secondary (i.e. due to another condition causing decreased heat production). Consider secondary causes in your differential diagnosis: sepsis, toxins, psychiatric illness, hypoglycemia, hypothyroidism, CNS dysfunction (e.g. stroke, head injury)

Hypothermia can occur even in normal temperatures

<u>Clinical classification:</u>

- **Mild:** normal mental status, shivering, normal VS including body temp 32.1-35 C (89.8-95 F)
- **Moderate/Severe:** altered LOC, no shivering, ↓BP, ↓HR, ↓RR, temp <32.1 C (89.8 F)

Primary Assessment

Look for associated injuries/illnesses

Consider Spinal Motion Restriction, if trauma is suspected

Remove wet garments & dry patient

Determine Severity of Exposure

Localized Cold Injury

- Dress injured area in clean cloths
- Do not rub skin
- Do not break blisters

- Monitor for signs of worsening hypothermia
- Consider transport to a Burn unit

Systemic Hypothermia

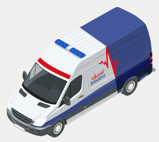
Is patient alert? — Yes / No

Is there pulse? — Yes / No

Yes:

- Cover patient with dry sheets or hypothermic blanket(s); apply hot packs if available
- Obtain 12-lead EKG
- Start IV, apply cardiac monitor
- Monitor VS, including EtCO2
- Do not allow patient to walk or stand
- Consider/treat causes of Altered Mental Status
- Transport to local facility

No pulse: Go to CPR CPG



Reassess

EFR | EMT | EMT ADVANCE | P | AP | EP

Heat Exposure

Hyperthermia increases heart demand and can cause dysrhythmias, acute MI, and heart failure. Monitor closely.

Primary Assessment

Move victim to a cool or shaded area
Remove clothing and loosen restrictive garments
Apply cold packs to the axilla, groin, and posterior neck
Determine severity of exposure

Heat Exhaustion

Elevated body temperature
Headache
Nausea /vomiting
Syncope
Heat cramps

PO Fluids
Water /Products with electrolytes

Crystalloid Fluid
if unable to tolerate PO
500mL IV/IO

Heat Stroke

Altered L.O.C
High body temperature
(Likely>40 C)
Hot dry skin
Hypotension
Possible seizure → Seizure guideline

Evaporative Cooling
- Fan the patient
- Mist skin with tepid H20

Crystalloid Fluid
500mL-1000mL IV/IO

Look for other causes of altered LOC using Altered Mental Status guideline

ECG & V/S

حكومــة دبــي
GOVERNMENT OF DUBAI

مؤسســـة دبـــي لخدمـــات الإسعـــاف
DUBAI CORPORATION FOR AMBULANCE SERVICES
إسعاف Ambulance

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Diving Emergencies

Complete primary survey
(Commence CPR if appropriate)

EFR

Consider diving buddy as possible patient also

Maintain ABC & Apply %100 O2 via Non-rebreather
Treat in Supine or Left Lateral Position

Complete Set of Vital Signs

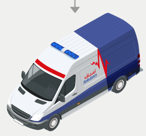
Consider ALS

NaCl (0.9%) 500 mL IV/IO

Continous Monitoring
ECG & SpO_2 & V/S

Transport dive computer and diving equipment with patient, if possible

Alert Dispatch



Treat associated symptoms as per the approperiate CPG
Penthrox is contraindicated

EFR	EMT	EMT ADVANCE	P	AP	EP

Sickle Cell Crisis - Adult

Primary Assessment

↓

Secondary Assessment

↓

Start O_2 15L/min via NRM

↓

Pain management required
- Yes → Go to Pain CPG (then return to flow)
- No ↓

Elevated temperature
- Yes → Go to Sepsis CPG
- No ↓

Signs of shock , Dehydration & unable to take oral intakes
- No → ETCO$_2$,SpO$_2$ & ECG & V/S monitor
- Yes ↓

Crystalloid Fluid 500mL-1000mL IV/IO

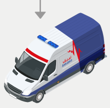
↓

Reassess

↓

ETCO$_2$,SpO$_2$ & ECG & V/S monitor

↓



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Adrenal Insufficiency – Adult

Primary Assessment

Secondary Assessment

- Recent illness or injury
  - No → Transport
  - Yes → Check blood glucose if < 70 mmHg → Go to Glycemic CPG
- SBP < 90 mmHg
  - No → Transport
  - Yes → **Hydrocortisone 100 mg IM/IV (in 100 mL NaCl)**
- Consider **Hydrocortisone 100 mg IM** if IV not available
- Reassess
- if SBP < 90 mmHg → **Normal Saline 250 mL-500mL Bolus**

Trauma Emergencies

SECTION 2

EFR | EMT | EMT ADVANCE | P | AP | EP

Primary Survey Trauma – Adult

Trauma

SCENE SIZE-UP
Standard Precautions Hazards, Number of Patients, Need for additional resources, Mechanism of Injury Consider pre - arrival information

Interruption of the Primary survey:
1.thescene becomes unsafe
2.you must treat exsanguinating hemorrhage
3.you must treat an airway obstruction
4.you must treat cardiac arrest

INITIAL ASSESSMENT
GENERAL IMPRESSION
Age, Sex, Weight, General Appearance, Position, Purposeful Movement, Obvious Injuries, Skin Color

Control catastrophic external haemorrhage

LOC (A-V-P-U)

A
Airway patent & protected

EFR
C-spine control

No

Jaw thrust

Suction
OPA

Yes

B
BREATHING (Present? Rate, Depth, Effort)

Consider
Oxygen therapy

Yes

C
CIRCULATION
(Radial/Carotid Present? Rate, Rhythm, Quality) Skin Color, Temperature, Moisture; Capillary Refill Has bleeding been controlled?

RAPID TRAUMA SURVEY OR FOCUSED EXAM

Clinical status decision

Life threatening

Serious not life threat

Non serious or life threat

Maximum time on scene for life-threatening trauma: ≤10 minutes

Request ALS

Go to appropriate CPG

Consider ALS

Go to Secondary Survey CPG

EFR | EMT | EMT ADVANCE | P | AP | EP

RAPID TRAUMA SURVEY

GCS for ≥ 4 years

Eye opening	
Spontaneously	4
To verbal stimuli	3
To painful stimuli	2
No response to pain	1
Best verbal response	
Orientated and converses	5
Confused and converses	4
Inappropriate words	3
Incomprehensible sounds	2
No response to pain	1
Best motor response	
Obeys verbal commands	6
Localises to stimuli	5
Withdraws to stimuli	4
Abnormal flexion to pain (decorticate)	3
Abnormal extension to pain (decerebrate)	2
No response to pain	1

GCS for < 4 years

Eye opening	
Spontaneously	4
To verbal stimuli	3
To painful stimuli	2
No response to pain	1
Best verbal response	
Appropriate words or social smile, fixes, follows	5
Cries but consolable; less than usual words	4
Persistently irritable	3
Moans to pain	2
No response to pain	1
Best motor response	
Spontaneous or obeys verbal commands	6
Localises to stimuli	5
Withdraws to stimuli	4
Abnormal flexion to pain (decorticate)	3
Abnormal extension to pain (decerebrate)	2
No response to pain	1

HEAD AND NECK
Neck Vein Distention? Tracheal Deviation?

↓

CHEST
Asymmetry; (w/Paradoxical Motion?), Contusions, Penetrations, TIC

↓

BREATH SOUNDS & HEART TONES
(Present? Equal? If unequal: percussion)

↓

ABDOMEN
(Contusions,Penetration/evisceration; Tenderness, Rigidity, Distention)

↓

PELVIS
Tenderness, Instability, Crepitation

↓

LOWER/UPPER EXTREMITIES
Obvious Swelling, Deformity

↓

POSTERIOR
Penetrations, Obvious Deformity

↓

VITAL SIGNS & GCS
BP, HR, RR, BSL, SpO_2

↓

PUPILS
Size, Equal, Reactive

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ONGOING EXAM

PATIENT HISTORY
Complete S.A.M.P.L.E. History if not already done

↓

LOC AVPU
Pupils -
If altered mental status: Finger-stick Glucose

↓

REPEAT INITIAL ASSESSMENT

↓

VITAL SIGNS
BP, HR, RR, BSL, SpO_2

↓

EXAMINE NECK
Obvious Wounds, Tenderness, Swelling , JVD? Tracheal Deviation?

↓

EXAMINE CHEST
Asymmetry (Paradoxical Motion?), Contusions, Penetrations, TIC

↓

BREATH SOUNDS & HEART TONES
(Present? Equal? (If unequal: Percussion)

↓

EXAMINE ABDOMEN
Contusions, Penetration /evisceration; Tenderness, Rigidity, Distension

↓

RECHECK IDENTIFIED INJURIES

↓

CHECK INTERVENTIONS
ET tube ,IVs, seal on chest wound ,decompression needle ,splints & dressings, impaled objects, position of pregnant patients

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Secondary Survey Trauma – Adult

Primary Survey

Generalized or unknown mechanism of injury

Yes → S.A.M.P.L.E. History

No → Examination of obvious injuries (Focused exam) → S.A.M.P.L.E. History

Monitor and record vital signs

GCS

Complete a head to toe survey DCAP-BLS, TIC

Identify positive findings and initiate care management

Go to appropriate CPG

Markers for multi-system trauma
Systolic BP < 90
Respiratory rate < 10 or > 29
Heart rate > 120
AVPU = V, P or U on scale
Mechanism of Injury

Deformities
Contusions
Abrasions
Penetrations
Burns
Lacerations
Swelling
Tenderness
Instability
Crepitus

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Traumatic Cardiac Arrest – Adult

EMS Unwitnessed Traumatic Arrest

Are patient's injuries incompatible with life (any of the following):

- Decapitation
- Incineration
- Body decomposition
- Rigor mortis

Yes → Go to Recognition of Death CPG

No ↓

Does patient meet ALL of following criteria:

- Victim of blunt or penetrating trauma
- No spontaneous movement
- No pupillary response
- Apneic and pulseless
- EKG showing asystole or PEA <40 bpm

Yes → Go to Recognition of Death CPG

No → Commence CPR

EMS Witnessed Traumatic Arrest → Commence CPR

Commence CPR → Consider ALS

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Traumatic Shock

INITIAL ASSESSMENT

RAPID TRAUMA SURVEY

Hemorrhagic Shock

Control bleeding
Oxygen
Fluid Management :-
maintain SBP more than 90 mmHg
Consider TXA 1g
(P)

None Hemorrhagic

Oxygen
Identify underlying cause and treat accordingly ,Ex :-
Tension Pneumothorax –Needle decompression .
Fluid Management:-
maintain SBP more than 90 mmHg

Pain Control

Maintain normothermia

Shock is defined as impaired tissue perfusion and may be manifested by any of the following:

- Altered mental status
- Tachycardia
- Poor skin perfusion
- Low blood pressure

Maintain a high index of suspicion. Traditional signs of shock may be absent early in the process.

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Burns – Adult

Stop the Burning Process

Airway Management
Maintain Ventilation
Manage Circulation

Electrical

ECG

Thermal

Fluid Management for 3rd-degree burns :-
Adult :- 2 ml LR x kg x % TBSA. 1/2 of the total fluid is provided in the first 8 hours after the burn injury .

Chemical

Rapid removal of the chemical :
Brush it away,
Remove clothess
or **Irrigating** with water and soap 10-15 minuts .

Cover burn with clean dressing

Prevent hypothermia

Pain Control

Burn (All degree) in following Area.
Should transport to hospital
Face
Hand
Flexion point
Pernium

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GOVERNMENT OF DUBAI

مؤسسة دبي لخدمات الإسعاف
DUBAI CORPORATION FOR AMBULANCE SERVICES
إسعاف Ambulance

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Head Injury – Adult

Adult GCS

Eye opening	
Spontaneously	4
To verbal stimuli	3
To painful stimuli	2
No response to pain	1
Best verbal response	
Orientated and converses	5
Confused and converses	4
Inappropriate words	3
Incomprehensible sounds	2
No response to pain	1
Best motor response	
Obeys verbal commands	6
Localises to stimuli	5
Withdraws to stimuli	4
Abnormal flexion to pain (decorticate)	3
Abnormal extension to pain (decerebrate)	2
No response to pain	1

SCENE SIZE-UP

↓

INITIAL ASSESSMENT
GENERAL IMPRESSION

↓

Life-threatening Bleeding

↓

V, P or U on AVPU → Yes → **Request ALS**

↓

AIRWAY
(WITH **C-SPINE** CONTROL)

↓

BREATHING Present?
Rate, Depth, Effort
= Consider Oxygen therapy

↓

CIRCULATION
Rate, Rhythm, Quality Skin Color, Temperature, Moisture; Capillary Refill

↓

RAPID TRAUMA SURVEY → Go to RAPID TRAUMA SURVEY CPG

↓

If GCS < 8,
Insert OPA/NPA , Advanced airway

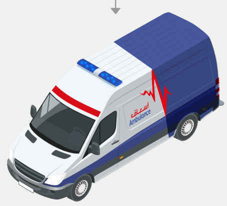
↓

Maintain SBP 110 -120 mmHg

↓

Initiate hyperventilation &
head up 30 degrees.
if s/s of herniation

↓



Signs of Herniation
Decreasing mental status.
Abnormal respiratory pattern .
Asymmetric/unreactive pupils.

Cushing s reflex (elevated BP, bradycardia, & irregular breathing/apnea).

Decorticate posturing (abnormal flexion)

Decerebrate posturing (extension)

Age Group	Normal Ventilation Rate	Hyperventilation Rate
Adult	8–10 breaths/minute (ETCO2 35–45)	20 breaths/minute ($ETCO_2$ about 30–35)
Children	15 breaths/minute ($ETCO_2$ 35–45)	25 breaths/minute ($ETCO_2$ about 30–35)
Infants	20 breaths/minute ($ETCO_2$ 35–45)	30 breaths/minute ($ETCO_2$ about 30–35)

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Spinal Injury Management

NEGATIVE

Mechanism of Injury

POSITIVE

Uncertain

Apply Manual Stabilization Until Exam Complete

Positive Mechanisms:

- High-Speed MVC
- Falls >Three Times Patient s Height
- Axial load
- Diving Accidents
- Penetrating Wound In or Near Spinal Column
- Sports Injuries to Head or Neck
- Unconscious Trauma Patient

Remove helmet (if worn)

SPINAL PAIN OR TENDERNESS?

yes

No

MOTOR AND SENSORY EXAM

ABNORMAL

Reliable Patient Is:

- Calm
- Cooperative
- Sober Alert
- Without Distracting Injuries

NORMAL

RELIABLE PATIENT?

yes

No

Unreliable Patient Has:

- Acute Stress Reaction
- Head/Brain Injury
- Altered Mental Status
- Intoxication with Drugs/Alcohol
- Other Distracting Injuries

Negative Spinal Injury: SMR NOT INDICATED

Positive Spinal Injury

Note: If any doubt exists...

Spinal Motion Restriction

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Limb Injury – Adult

INITIAL ASSESSMENT

↓

RAPID TRAUMA SURVEY

↓

Expose & Support injured limb with check PMS

↓

Pain Control → Go to appropriate CPG

↓

Dress open wounds as needed

↓

Apply appropriate splinting device with check PMS

↓

Special Considration

PELVIC TRAUMA

-Pelvic binder

-Fluid Management:-

1500-2000ml

Fracture Femur:-

Fluid Management:- 1000-1500 ml

Mid shaft fracture A pply traction splint.

Anterior Shoulder Dislocation :-

Gentle external rotation of the humerus.

(AP,EP)

Dislocation **Patella Trauma :-**

Reduce dislocation and apply splint

(AP,EP)

Fracture Patella Trauma :-

Splint in position found.

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Crush Injury

Patient trapped

Maintain AcBC

Oxygen therapy

AcBC
Airway
cervical spine
Breathing
Circulation

Co-ordinate with rescue personnel on release timing

Significant compression force maintained

No → Apply standard trauma care during and post extrication

Yes ↓

IV access Large bore x 2

Consider pain relief → Go to Pain Mgt. CPG

Consider tourniquet use , NaCl (0.9%) 20 mL/Kg IV/IO

ECG and SpO_2 monitoring

If possible commence IV fluids prior to release

Prepare all required patient carrying devices and have on standby following extrication

Apply standard trauma care during and post extrication

Go to appropriate CPG

Pediatrics Emergencies

SECTION 3

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Primary Assessment Medical – Paediatric (≤ 13Years)

Medical Complaints

The primary assessment is focused on establishing the patient's clinical status and only applying interventions when they are essential to maintain life. It should be completed within one minute of arrival on scene.

SCENE SIZE UP

Consider pre - arrival information

General Impression
(general appearance ,age ,sex ,aggressive, etc..)

AVPU

- Unconscious unresponsive → Go to CPR CPG
- Full Conscious → Chief Complaints
- Alter conscious level → Go to AMS CPG

Chief Complaints

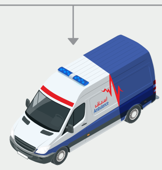
Airway
patent or not/open by Heat tilt-chin lift or jaw thrust is suspected cervical injury

Breathing
Look, Listen, Feel the chest, give oxygen if spo2:less than %94

Circulation
pulse (Present or absent, weak, fast or regular) skin(cold ,rash,pale,cyanose,temp),

Priority and Triage

Secondary Assessment - Paediatric



Normal ranges (ICTS)

Age	Pulse
0-3 months	85 – 205
3 months - 2 years	100 – 190
2-10 years	60 – 140
>10 years	60 – 100

Normal ranges (ICTS)

Age	RR
Infant	30 – 60
Toddler	24 – 40
Preschooler	22 – 34
School-aged child	18 – 30
Adolescent	12 – 16

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Secondary Assessment – Paediatric (≤ 13 years)

Primary assessment

Normal ranges (ICTS)

Age	Pulse
0-3 months	85 – 205
3 months - 2 years	100 – 190
2-10 years	60 – 140
>10 years	60 – 100

Normal ranges (ICTS)

Age	RR
Infant	30 – 60
Toddler	24 – 40
Preschooler	22 – 34
School-aged child	18 – 30
Adolescent	12 – 16

SAMPLE history

HEAD TO TOE EXAMINATION

IV Acess / IV Fluid
Medication intervention

VITAL SIGNS
Bp ,HR ,RR,Temp,Sugar,Pupils Size,Skin ,Ecg

Repeate Vitals Sign every 5 minutes for unstable patient and every 15m for stable pt

Reassess

Children and adolescents should always be examined with a chaperone (usually a parent) where possible

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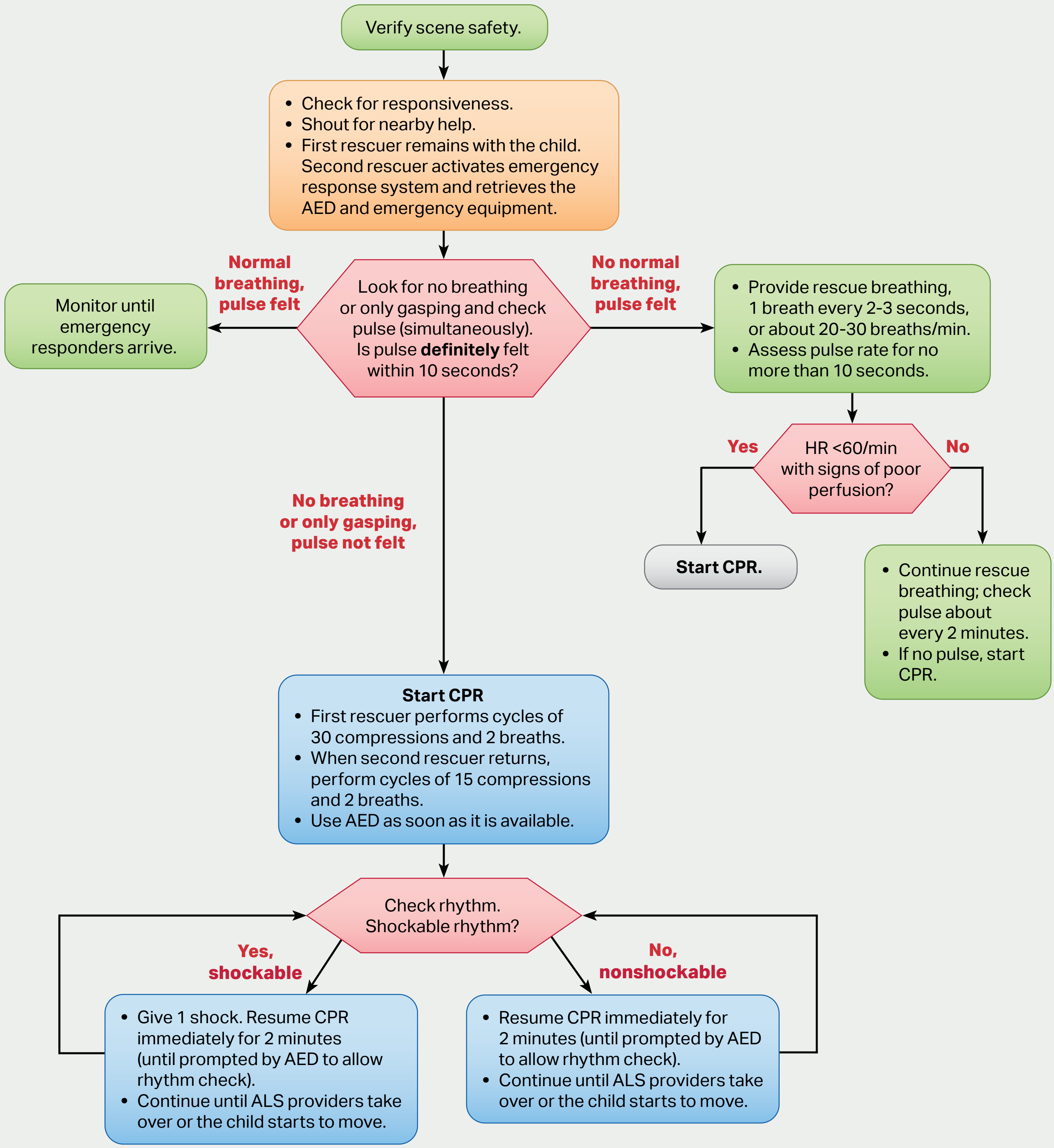
Basic Life Support–Paediatric (≤ 13 years) 2 or More Rescuers



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Paediatric Cardiac Arrest (≤ 13 years)

1 Start CPR
- Begin bag-mask ventilation and give oxygen
- Attach monitor/defibrillator

Rhythm shockable?

Yes →

2 VF/pVT

3 Shock

4 CPR 2 min
IV/IO access

Rhythm shockable? (No → 12)

Yes

5 Shock

6 CPR 2 min
- **Epinephrine** every 3-5 min
- Consider advanced airway

Rhythm shockable? (No → 12)

Yes

7 Shock

8 CPR 2 min
- **Amiodarone** or **lidocaine**
- Treat reversible causes

No →

9 Asystole/PEA

Epinephrine ASAP

10 CPR 2 min
- IV/IO access
- **Epinephrine** every 3-5 min
- Consider advanced airway and capnography

Rhythm shockable? (Yes → Go to 7.)

No

11 CPR 2 min
Treat reversible causes

Rhythm shockable?

No → 12

Yes → Go to 7.

12
- If no signs of return of spontaneous circulation (ROSC), go to **10**
- If ROSC, go to Post–Cardiac Arrest Care checklist

Go to **7**.

CPR Quality
- Push hard (≥⅓ of anteroposterior diameter of chest) and fast (100-120/min) and allow complete chest recoil
- Minimize interruptions in compressions
- Change compressor every 2 minutes, or sooner if fatigued
- If no advanced airway, 15:2 compression-ventilation ratio
- If advanced airway, provide continuous compressions and give a breath every 2-3 seconds

Shock Energy for Defibrillation
- First shock 2 J/kg
- Second shock 4 J/kg
- Subsequent shocks ≥4 J/kg, maximum 10 J/kg or adult dose

Drug Therapy
- **Epinephrine IV/IO dose:**
  0.01 mg/kg (0.1 mL/kg of the 0.1 mg/mL concentration).
  Max dose 1 mg.
  Repeat every 3-5 minutes.
  If no IV/IO access, may give endotracheal dose: 0.1 mg/kg (0.1 mL/kg of the 1 mg/mL concentration).
- **Amiodarone IV/IO dose:**
  5 mg/kg bolus during cardiac arrest. May repeat up to 3 total doses for refractory VF/pulseless VT
  or
  Lidocaine IV/IO dose:
  Initial: 1 mg/kg loading dose

Advanced Airway
- Endotracheal intubation or supraglottic advanced airway
- Waveform capnography or capnometry to confirm and monitor ET tube placement

Reversible Causes
- **H**ypovolemia
- **H**ypoxia
- **H**ydrogen ion (acidosis)
- **H**ypoglycemia
- **H**ypo-/hyperkalemia
- **H**ypothermia
- **T**ension pneumothorax
- **T**amponade, cardiac
- **T**oxins
- **T**hrombosis, pulmonary
- **T**hrombosis, coronary

ROSC - Postresuscitation Care

Optimize Ventilation and Oxygenation
. Titrate Fio2 to maintain oxyhemoglobin saturation 94%-99%; if possible, wan Fio2 if saturation is 100%
. Consider advance airway placement and waveform capnography

Asses for and Treat Persistent shock
- Identify, treat contributing factors.*
- Consider 20 mL/kg IV/IO boluses of isotonic crystalliod. Consider smaller blouses (eg,10 mL/kg) if poor cardiac function suspected.
- Consider the need for inotropic and/ or vasopressor support for fluid-refractory shock.

***Possible Contributing Factors**
Hypovolemia
Hypoxia
Hydrogen ion (acidosis)
Hypoglycemia
Hypo-/hyperkalemia
Hypothermia
Tension pnuemothorax
Tamponade, cardiac
Toxins
Thrombosis, pulmonary
Thrombosis, coronary
Trauma

Hypotensive shock
- Epinephrine
- Dopamine
- Norepinephrine

Normotensive Shock
- Dobutamine
- Dopamine
- Epinephrine
- Milrinone

- Monitor for and treat agitation and seizures
- Monitor for and treat hypoglycemia
- Assess blood gas, serum electrolytes, calcium from cardiac arrest, consider therapeutic hypothermia (32°C-34°C)
- Consider consultation and patient transport to tertiary care center.

Estimation of Maintenance Fluid Requirements

Infants <10 kg:
4mL/kg per hour

Example: For an 8-kg infant, estimated maintenance fluid rate.
= 4mL/kg per hour x 8kg
= 32mL per hour

Children 10-20 kg:
4mL/kg per hour for the first 10kg
+ 2mL/kg per hour for each kg above 10 kg.

Example: For a 15-kg child, estimated maintenance fluid rate
= (4mL/kg per hour x10kg) + (2mL/kg per hour x 5kg)
=40mL/hour + 10mL/hour
=50mL/hour

Children >20 kg:
4mL/kg per hour for the first 10kg + 2mL/kg per hour kg 11-20 + 1mL/kg per hour for each kg above 20 kg.

Example: For a 28-kg child, estimated maintenance fluid rate
= (4mL/kg per hour x 10kg) + (2mL/kg per hour x 10 kg) +(1 mL/kg per hour x 8kg)
=40 mL per hour + 20 mL per hour + 8 mL per hour
= 68 mL per hour

Following initial stabilization, adjust the rate and composition of intravenous fluids based on the patient's clinical condition and state of hydration. In general, provide a continuous infusion of a dextrose containing solution for infants. Avoid hypotonic solutions in critically ill children; for most patients use isotonic fluid such as normal saline (0.9% NaCl) or lactated Ringer's solution with or without dextrose, based on the child's clinical status.h

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Symptomatic Bradycardia – Paediatric (≤ 13 years)

Patient with bradycardia

Cardiopulmonary compromise?
- Acutely altered mental status
- Signs of shock
- Hypotension

No →
- Support ABCs
- Consider oxygen
- Observe
- 12-Lead ECG
- Identify and treat underlying causes

Yes →

Assessment and support
- Maintain patent airway
- Assist breathing with positive pressure ventilation and oxygen as necessary
- Cardiac monitor to identify rhythm; monitor pulse, BP, and oximetry

Start CPR if HR <60/min despite oxygenation and ventilation.

Bradycardia persists?

No → Support ABCs / Consider oxygen / Observe / 12-Lead ECG / Identify and treat underlying causes

Yes →
- Continue CPR if HR <60/min
- IV/IO access
- **Epinephrine**
- **Atropine** for increased vagal tone or primary AV block
- Consider transthoracic/ transvenous pacing
- Identify and treat underlying causes

Check pulse every 2 minutes. **Pulse present?**

Yes → Bradycardia persists?

No → Go to **Pediatric Cardiac Arrest Algorithm.**

Doses/Details

Epinephrine IV/IO dose:
0.01 mg/kg (0.1 mL/kg of the 0.1 mg/mL concentration). Repeat every 3-5 minutes. If IV/IO access not available but endotracheal (ET) tube in place, may give ET dose: 0.1 mg/kg (0.1 mL/kg of the 1 mg/mL concentration).

Atropine IV/IO dose:
0.02 mg/kg. May repeat once. Minimum dose 0.1 mg and maximum single dose 0.5 mg.

Possible Causes
- Hypothermia
- Hypoxia
- Medications

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Paediatric Tachycardia (≤ 13 years)

Pediatric Tachycardia With a Pulse Algorithm

Initial assessment and support
- Maintain patent airway; assist breathing as necessary
- Administer oxygen
- Cardiac monitor to identify rhythm; monitor pulse, blood pressure, and oximetry
- IV/IO access
- 12-Lead ECG if available

Evaluate rhythm with 12-lead ECG or monitor.

Probable sinus tachycardia *if*
- P waves present/normal
- Variable RR interval
- Infant rate usually <220/min
- Child rate usually <180/min

→ **Search for and treat cause.**

Cardiopulmonary compromise?
- Acutely altered mental status
- Signs of shock
- Hypotension

Yes → **Evaluate QRS duration.**

- **Narrow (≤0.09 sec)** → **Probable supraventricular tachycardia**
  - P waves absent/abnormal
  - RR interval not variable
  - Infant rate usually ≥220/min
  - Child rate usually ≥180/min
  - History of abrupt rate change

  → • If IV/IO access is present, **adenosine**
  or
  • If IV/IO access is not available, or if adenosine is ineffective, perform synchronized cardioversion
- **Wide (>0.09 sec)** → **Possible ventricular tachycardia** → **Synchronized cardioversion**
  Expert consultation is advised before additional drug therapies.

No → **Evaluate QRS duration.**

- **Narrow (≤0.09 sec)** → **Probable supraventricular tachycardia**
  - P waves absent/abnormal
  - RR interval not variable
  - Infant rate usually ≥220/min
  - Child rate usually ≥180/min
  - History of abrupt rate change

  → **Consider vagal maneuvers.** → If IV/IO access is present, give **adenosine.**
- **Wide (>0.09 sec)** → **Possible ventricular tachycardia** → If rhythm is **regular** and QRS **monomorphic**, consider **adenosine**. → **Expert consultation is recommended.**

Doses/Details

Synchronized cardioversion
Begin with 0.5-1 J/kg; if not effective, increase to 2 J/kg. Sedate if needed, but don't delay cardioversion.

Drug Therapy

Adenosine IV/IO dose
- First dose: 0.1 mg/kg rapid bolus (maximum: 6 mg)
- Second dose: 0.2 mg/kg rapid bolus (maximum second dose: 12 mg)

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Allergic Reaction/Anaphylaxis – Paediatric (≤ 13 years)

Primary Assessment

O_2 therapy

Mild

Monitor reaction

Antihistamines Syrup

Moderate

Diphenhydramine
1 mg /kg IM Max 50 mg

+

Hydrocortisone
< 1 yr 25 mg IM
1-5 yrs 50 mg IM
> 5 yrs 100 mg IM

If bronchospasm consider nebuliser

Salbutamol
< 5 yrs: 2.5 mg
>5 yrs: 5 mg

Reassess

Deteriorates

Yes → Repeat Epinephrine (auto injector)

No → ECG & SpO_2 monitor

Severe/ Anaphylaxis

Epinephrine (auto injector)
<6 yrs : 0.15 mg
>6 yrs :0.3 mg

+

Crystalloid fluids 10 ml/ kg

+

Diphenhydramine
1 mg /kg IM Max 50 mg

+

Hydrocortisone
< 1 yr 25 mg IM
1-5 yrs 50 mg IM
> 5 yrs 100 mg IM

If bronchospasm consider nebuliser

Salbutamol
< 5 yrs: 2.5 mg
>5 yrs: 5 mg

Reassess

Deteriorates

Repeat Epinephrine (auto injector)
<6 yrs : 0.15 mg
>6 yrs :0.3 mg

call TeleEMS

ECG & SpO_2 monitor

Mild
Urticaria

Moderate
Mild symptoms + angio oedema
simple bronchospasm

Severe/ anaphylaxis
Moderate symptoms + ABC compromise
Abdominal pain
V°/ D°, Dizziness or syncope

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Pain Management – Paediatric (≤ 13years)

Scene Size Up → **Primary assessment**

Analogue or Visual Pain Scale
0 = no pain....10 = unbearable

Adequate relief of pain — Yes or best achievable → Go to appropriate CPG

No ↓

Implement pharmacology strategy at appropriate level on the pain ladder

Mild pain
(1 to 3 on pain scale)

Paracetamol PR
5 years>125 mg
5 years<250 mg

EMT

Paracetamol IV not recommended for less than 33Kg

Moderate pain
(4 to 6 on pain scale)

Paracetamol 15mg/kg PR or IV

EMT

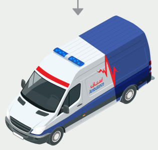
Decisions to give analgesia must be based on clinical assessment and not directly on a linear scale

Severe pain
(7 or more on pain scale)

Morphine sulphate 0.1 MG/kg IV/IM

Practitioners, depending on his/her scope of practice, may make a clinical judgment and commence pain relief on a higher rung of the pain ladder.

Reassess



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Shock – Paediatric (≤ 13years)

General Signs of shock
- Altered mental status
- Delayed/flash capillary refill
- Weak or decreased pulses
- Tachycardia
- Elevated RR
- Hypoxemia
- Hypotension for age
- Decreased urine output

Primary Assessment

Place 1 or 2 large bore IV s 18g
(consider IO after two unsuccessful attempts)

Consider etiology of shock state

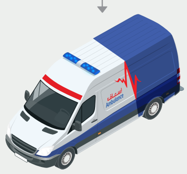
Anaphylactic / Septic / Hypovolemic / Neurogenic
- **Crystalloid Fluid bolus:** 10 mL/kg

Obstructive
- High flow O_2 via NRB
- Evaluate lung sounds
- **Decompress tension pneumothorax**

Cardiogenic
- Evaluate lung sounds
- **Crystalloid Fluid bolus:** 10 ml/kg if lungs sounds clear bilaterally

Reassess

if persistent shock repeat Crystalloid Fluid bolus 10 ml/kg



Hypovolemic
Hemorrhage – trauma
or
medical (ex. GI bleed)
Vomiting/Diarrhea
Burns

Cardiogenic
Heart Failure
Arrhythmias
Cardiomyopathy
Valvular Disease

Obstructive
Tension pneumothorax
Hemopneumothorax
Massive PE
Cardiac tamponade

Distributive
Septicemia
look for source & fever
may have a low $EtCO_2$
Anaphylaxis
often warm/flush skin
Neurogenic
usually h/o trauma
often bradycardic

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Sepsis – Paediatric (≤ 13years)

Patient unwell →

Signs of Systemic Inflammatory Response Syndrome (SIRS)

- Temperature < 36 or > 38.5oC
- Heart rate - tachycardia
- Respiratory rate - tachypnoea
- Acutely confused
- Glucose > 140 (not diabetic)

Has the patient two or more signs (SIRS)

No → (to transport)

Yes ↓

Could this be a severe infection?
For example

- Pneumonia
- Meningitis/ meningococcal disease
- UTI
- Abdominal pain or distension
- Indwelling medical device
- Cellulitis/ septic arthritis/ infected wound
- Chemotherapy < 6 weeks
- Recent organ transplant
- On immune-suppressant medication

No → (to transport)

Yes ↓

ECG, SpO_2 & BP monitoring

↓

Oxygen therapy

↓

Signs of poor perfusion

Yes → NaCl (0.9%) 10 mL/Kg IV/IO
Repeat prn

No ↓

If temperature > 38.5ºC consider

Paracetamol PR
<5 years 125 mg
>5 years 250 mg
Or
Paracetamol inj
(>35kg) 10mg /kg IV

If meningitis suspected ensure appropriate PPE is worn; Mask and goggles

Signs of inadequate perfusion

A: (not directly affected)
B: Increased respiratory rate (without increased effort)
C: Tachycardia
Diminished/absent peripheral pulses Delayed capillary refill
D: Irritability/ confusion / ALoC
E: Cool extremities, mottling

Normal ranges (ICTS)

Age	Pulse
0-3 months	85 – 205
3 months - 2 years	100 – 190
2-10 years	60 – 140
>10 years	60 – 100

Normal ranges (ICTS)

Age	RR
Infant	30 – 60
Toddler	24 – 40
Preschooler	22 – 34
School-aged child	18 – 30
Adolescent	12 – 16

EFR	EMT	EMT ADVANCE	P	AP	EP

Altered Mental Status Paediatric (≤ 13 years)

Potential Causes& Corresponding Guidelines

Alcohol	→	Drug Overdose
Epilepsy	→	Seizure
Insulin	→	Diabetic Emergency
Opiates	→	Drug Overdose
Uremia	→	N/A (dialysis needed)
Trauma	→	Traumatic Cardiac Arrest Traumatic Shock Head Injury
Temperature	→	Hypo-/Hyperthermia
Infection	→	Shock
Psychosis	→	Behavioral Emergency
Stroke	→	Stroke
Seizure	→	Seizure

Primary Assessment ABC Assessment

↓

Upper airway or respiratory compromise? — Yes → **Airway Management guideline(s)**

No ↓

Signs of opiate/opioid overdose? — Yes → **Drug Overdose guideline**

No ↓

Signs of shock? — Yes → **Shock guideline**

No ↓

Obvious signs of trauma? → **Routine Trauma Care & appropriate Trauma guideline**

No ↓

BSL < 70 mg/dl — Yes → **Diabetic Emergency guideline**

No ↓

History of agitation or erratic behavior? — Yes → **Behavioural emergency guidline**

No ↓

Recent or current seizure activity? — Yes → **Seizure guideline**

No ↓

Positive Cincinnati Prehospital Stroke Scale? — Yes → **Stroke guideline**

No ↓

Consider other causes of AMS:
ex. encephalopathy, dysrhythmia, hypoxia, hypercapnea, CO poisoning, toxiciology

Shock is defined as impaired tissue perfusion and may be manifested by any of the following :

- Altered mental status
- Tachycardia
- Poor skin perfusion
- Low blood pressure

Maintain a high index of suspicion . Traditional signs of shock may be absent early in the process

Contact **TeleEMS** for additional orders or consultation

EFR | EMT | EMT ADVANCE | P | AP | EP

Seizure/Convulsion – Paediatric (≤ 13years)

Consider other causes of seizures
Meningitis
Head injury Hypoglycaemia
Fev er / Febrile Convulsion
Poisons

Primary Assessment & neurological examination

Protect from harm

Oxygen therapy

Seizure status

Seizing currently

Request ALS

- IM midazolam 0.2 mg/kg (max 10 mg)
- Rectal diazepam 0.5 mg/kg (max 20 mg)
- Diazepam 0.2 mg/kg IM /IV (max 10 mg)

Support head

Check blood glucose

Blood glucose < 70 mg/dL
Yes: Go to Glycaemic Emergency CPG
No

Reassess

Still seizing
No → Post seizure
Yes

Post seizure

Alert
Yes → Check blood glucose
No

Recovery position

Airway management

Check blood glucose

Blood glucose < 75mg/dlL
Yes: Go to Glycaemic Emergency CPG
No

Reassess

EFR | EMT | EMT ADVANCE | P | AP | EP

Airway Management – Paediatric (≤ 13years)

Compromise Airway

Scene Size Up

Primary Assessment

OPEN AIRWAY,CLEAR ANY FORIENBODY OR SCUTION IT IF ANY VOMITUS OR BLOOD
(Headtilt -chin left if no suspected cervical injury, Jaw-thrust if suspected cervical injury)

Oxygen by NRB 10-15 L
BVM+/- Adjunct Airway

Adjunct Airway
OPA/NPA

If not maintained the ventilation
Call back up or TELE-EMS

ADVANCE AIRWAY
(Endotracheal Tube/Laryngeal Tube)

Successful
(Chest rise/ Air entry/Absent Gastric sound)

Yes → connect to oxygen 100%, Or Ventilator

No ↓

2nd attempt
(Endotracheal Tube/Laryngeal Tube)

Successful

Yes → Give 1 breath every 5-6 second
+attach capnography

Capnography Etco$_2$ 35-45 mmHg

No ↓

Revert to basic airway
BVM+/- Adjunct Airway

Reassess

EFR	EMT	EMT ADVANCE	P	AP	EP

Foreign Body Airway Obstruction – Paediatric (≤ 13years)

Scene Size Up → **Primary assessment**

Is patient conscious?

Yes:

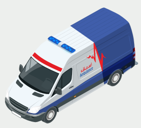
Are there signs of severe airway obstruction?
(e.g. stridor, <u>Ineffective</u> cough, or severe respiratory distress)

Yes:

Infant <1 year Give continuous 5 Back slap and chest thrust
>1 year 5 Black slap and abdominal thrust until FB is expelled or patient becomes unconscious

NO:

- Encourage forceful coughing
- Observe for decompensation
- Do not interfere with patient spontaneous coughing



NO:

Begin CPR
(chest compressions)

- Look into mouth before ventilating
- Remove FB with finger sweep <u>only</u>
- if FB is visualized
- Suction airway as needed

- Initiate ventilations via BVM with 15L/min Oxygen
- Continue chest compressions with ventilations

EFR	EMT	EMT ADVANCE	P	AP	EP

Croup – Paediatric (≤ 13 years)

Scene Size Up → Primary assessment

Oxygen therapy

Exclude Foreign Body Obstruction as Cause

Attempt to keep child calm, avoid undue distress
Allow for position of comfort

Mild

Dexamethasone 0.6mg/kg /PO

Mild Croup:
Occasional barking cough
Little or no stridor at rest
Mild or Absent retraction
Moderate :
Frequent barking cough
Audible Stridor at rest
Retraction at rest
good air entry on auscultation of lung
Little or no agitation
Sever Croup:
Frequent barking cough
Marked Retraction
decreased Air entry by auscultation of lung
Spo2 <92%
Significant Agitation
Decrease level of consciousness

Moderate

Request ALS

Dexamethasone 0.6mg/kg /PO

Not improved

Nebulized Racemic Epinephrine (2.25%)0.05ml/kg Max 0.5ml

Severe

Request ALS

Dexamethasone 0.6mg/kg /PO
+
Nebulized Racemic Epinephrine (2.25%)0.05ml/kg Max 0.5ml

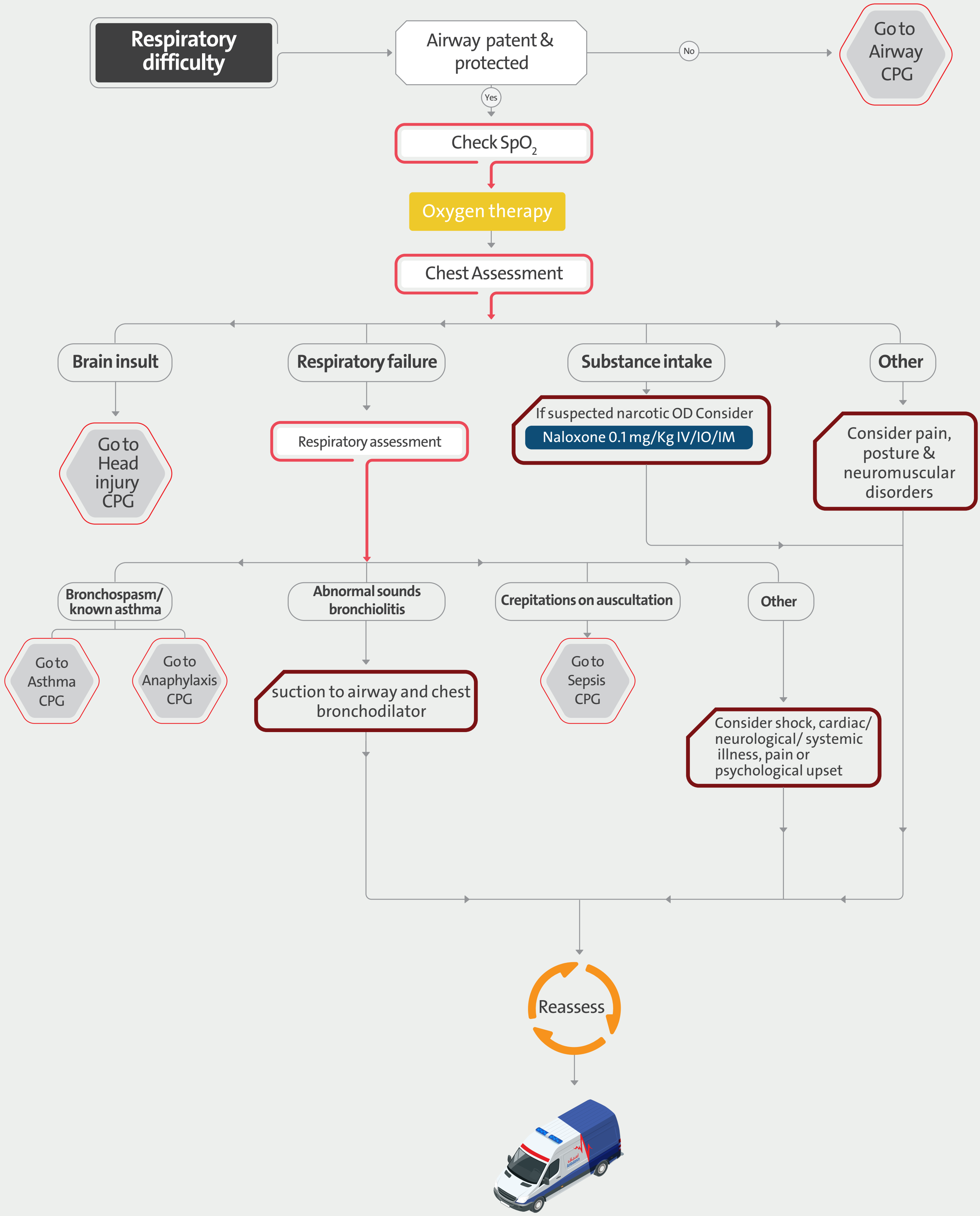
Not improved

Consider Advanced Airway Management

Reassess

EFR | EMT | EMT ADVANCE | P | AP | EP

Difficulty Of Breathing – Paediatric (≤ 13 years)



EFR	EMT	EMT ADVANCE	P	AP	EP

Asthma – Paediatric (≤ 13 years)

SCENE SIZE UP

↓

Primary Assessment

↓

Chest examination

↓

Oxygen therapy (if SPO2 < 94%) by NRB

↓

ECG & SpO_2 and V/S monitoring

↓

Mild Asthma

< 5 years Salbutamol 2.5 mg NEB
≥ 5 years Salbutamol 5 mg NEB

↓

Resolved/ improved — Yes → ambulance

No ↓

Moderate Asthma

< 5 years Salbutamol 2.5 mg NEB
≥ 5 years Salbutamol 5 mg NEB

+

Dexamethasone 0.6mg / kg PO

or

Hydrocortisone
< 1 yr 25 mg IM
1-5 yrs 50 mg IM
> 5 yrs 100 mg IM

↓

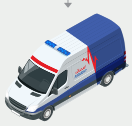
Severe and Life-threatening Asthma

Request ALS and call TeleEMS

↓

MG sulphate 40 mg/kg IV inj — AP

↓



Mild
Wheezes only on auscultation
Dry cough at night or after exercise
Spo2 >95%.
Moderate:
Able to talk in sentences
Spo2>92%
PEF>50%
HR<140/m in children 1-5 years
HR <125/m > 5 years
RR <40/m (1-5 years)
<30/m (> 5 years)
Sever Asthma
Cant complete sentences in one breath
Spo2<92%
HR
140/m (1-5 years)
125/m (> 5years)
RR
>40/m (1-5 years)
>30%(>5 years)
Life threatening
Exhaustion
Hypotension
Cyanosis
Silent chest
Confusion
PEF <33%
Spo2<92

EFR | EMT | EMT ADVANCE | P | AP | EP

Pyrexia – Paediatric (≤ 13 years)

Child with elevated temperature
> 38°C

Convulsion — Yes → Go to Convulsion CPG

Remove/protect from hot environment
(providing it is safe to do so)

Alert

- No → Recovery position (maintain airway)
  Assess vital signs/RBS
  → Query severe Sepsis
  - Yes → Go to Septic Shock CPG
  - No → Cool patient
- Yes → Cool patient

Paracetamol PR
<5 years125 mg
>5 years250 mg
Or
Paracetamol inj
(>35kg) 10mg /kg IV

SpO_2 & ECG monitor

EFR | EMT | EMT ADVANCE | P | AP | EP

Glycaemic Emergency – Paediatric (≤ 13 years)

When unable to obtain a reading then assume CBG is low if patient is altered & takes insulin

Primary Assessment

Assess Blood Glucose

< 60 mg/ dl

Glucose gel
<8 years 5-10 g Buccal
≥ 8 years 10-20 g Buccal
or
Glucagon
1 - 8 years 0.5 mg IM
> 8 years 1 mg IM
or
Dextrose
10% 2-5 mL/Kg

- Repeat CBG if altered LOC persists and treat accordingly
- Look for other causes of altered LOC if maximal doses of dextrose do not improve patient s mental status

call TeleEMS
if further orders are needed.

CBG > 250mg/dL and signs of ketoacidosis

NaCl (0.9%) 10 mL/Kg IV/IO bolus

12-Lead EKG

Reassess patient for fluid overload & cerebral edema

Additional **fluid** for extended transports as ordered by **TeleEMS**

Check for presence of an insulin pump; turn off or remove if present
Glucagon not indicated for hypoglycaemic non-diabetic paediatric patients

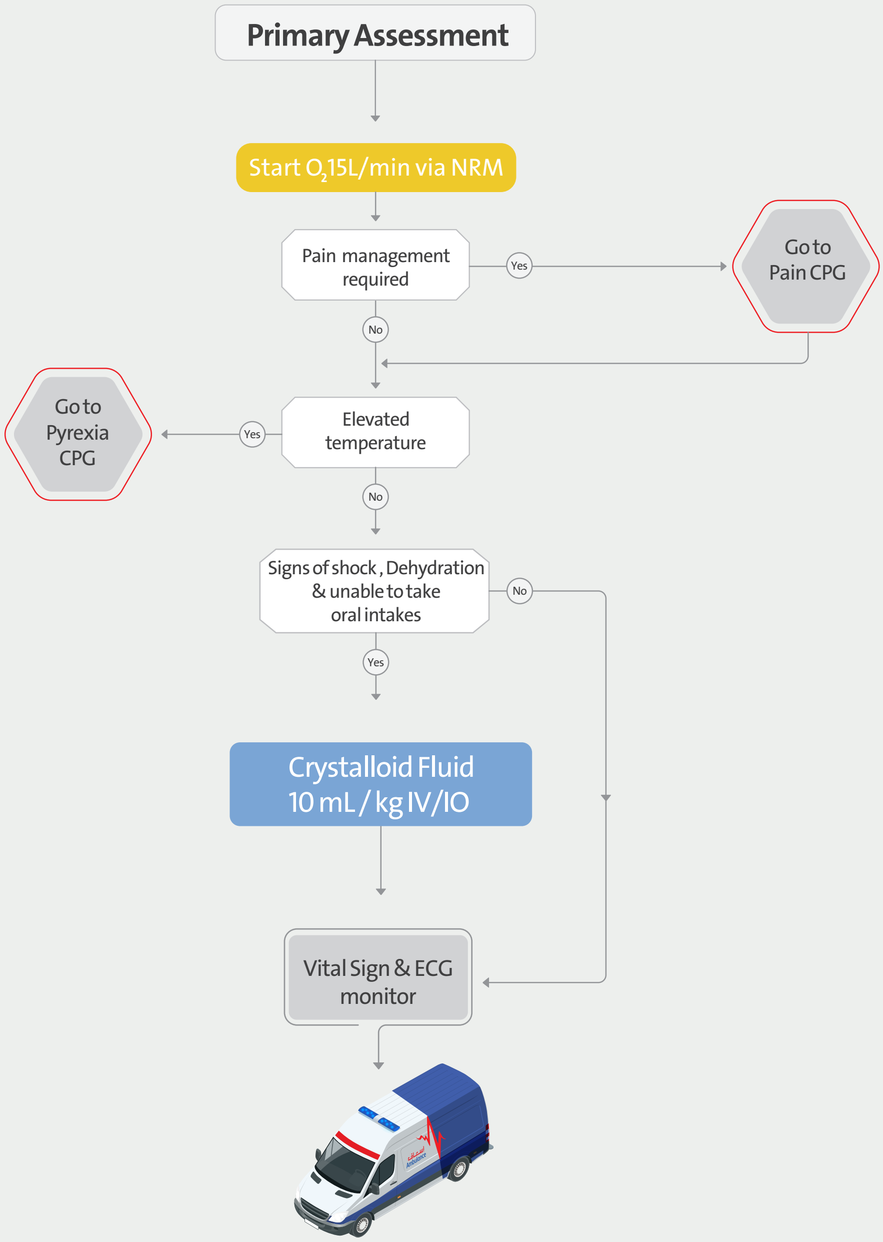
Signs of Ketoacidosis

- Ill appearance
- Dyspnea
- Deep rapid (Kussmaul) respirations
- Tachycardia +/ - weak pulses
- Nausea or vomiting
- Abdominal pain or cramping
- Headache or double vision
- Altered LOC
- Fruity/acetone breath or odor
- Muscle wasting or weight loss
- Flushed/dry skin, skin tenting
- Dry mucous membranes

Remember other causes of ketoacidosis – ex . starvation & alcohol

EFR	EMT	EMT ADVANCE	P	AP	EP

Sickle Cell Crisis – Paediatric (≤ 13years)



EFR | EMT | EMT ADVANCE | P | AP | EP

Adrenal Insufficiency – Paediatric (≤ 13 years)

Diagnosed with Addison's disease or Adrenal insufficiency

Recent illness or injury
- No → NaCl (0.9%) 10 mL/Kg IV
- Yes → Check blood glucose

Poor perfusion
- No → NaCl (0.9%) 10 mL/Kg IV
- Yes → Hydrocortisone

Hydrocortisone
< 1 yr 25 mg IM
1-5 yrs 50 mg IM
> 5 yrs 100 mg IM

Reassess

NaCl (0.9%) 10 mL/Kg IV

EFR | EMT | EMT ADVANCE | P | AP | EP

Primary Survey Trauma Paediatric (≤ 13 years)

SCENE SIZE-UP
Standard Precautions Hazards, Number of Patients, Need for additional resources, Mechanism of Injury Consider pre - arrival information

Interruption of the Primary survey:
1.the scene becomes unsafe
2.you must treat exsanguinating hemorrhage
3.you must treat an airway obstruction
4.you must treat cardiac arrest

INITIAL ASSESSMENT
GENERAL IMPRESSION
Age, Sex, Weight, General Appearance, Position, Purposeful Movement, Obvious Injuries, Skin Color

Control catastrophic external haemorrhage

LOC (A-V-P-U)

A
Airway patent & protected

EFR
C-spine control

No
Jaw thrust = Suction OPA

Yes

B
BREATHING (Present? Rate, Depth, Effort) = Consider Oxygen therapy

Yes

C
CIRCULATION
(Radial/Carotid Present? Rate, Rhythm, Quality) Skin Color, Temperature, Moisture; Capillary Refill Has bleeding been controlled?

RAPID TRAUMA SURVEY OR FOCUSED EXAM

Clinical status decision

Life threatening
Maximum time on scene for life-threatening trauma: ≤ 10 minutes
Request ALS → Go to appropriate CPG

Serious not life threat
Consider ALS → Go to appropriate CPG / Go to Secondary Survey CPG

Non serious or life threat
Go to Secondary Survey CPG

EFR | EMT | EMT ADVANCE | P | AP | EP

RAPID TRAUMA SURVEY Paediatric (≤ 13 years)

GCS for ≥ 4 years

Eye opening	
Spontaneously	4
To verbal stimuli	3
To painful stimuli	2
No response to pain	1
Best verbal response	
Orientated and converses	5
Confused and converses	4
Inappropriate words	3
Incomprehensible sounds	2
No response to pain	1
Best motor response	
Obeys verbal commands	6
Localises to stimuli	5
Withdraws to stimuli	4
Abnormal flexion to pain (decorticate)	3
Abnormal extension to pain (decerebrate)	2
No response to pain	1

GCS for < 4 years

Eye opening	
Spontaneously	4
To verbal stimuli	3
To painful stimuli	2
No response to pain	1
Best verbal response	
Appropriate words or social smile, fixes, follows	5
Cries but consolable; less than usual words	4
Persistently irritable	3
Moans to pain	2
No response to pain	1
Best motor response	
Spontaneous or obeys verbal commands	6
Localises to stimuli	5
Withdraws to stimuli	4
Abnormal flexion to pain (decorticate)	3
Abnormal extension to pain (decerebrate)	2
No response to pain	1

HEAD AND NECK
Neck Vein Distention? Tracheal Deviation?

↓

CHEST
Asymmetry; (w/Paradoxical Motion?), Contusions, Penetrations, TIC

↓

BREATH SOUNDS & HEART TONES
(Present? Equal? If unequal: percussion)

↓

ABDOMEN
(Contusions, Penetration/evisceration; Tenderness, Rigidity, Distention)

↓

PELVIS
Tenderness, Instability, Crepitation

↓

LOWER/UPPER EXTREMITIES
Obvious Swelling, Deformity

↓

POSTERIOR
Penetrations, Obvious Deformity

↓

VITAL SIGNS & GCS
BP, HR, RR, BSL, SpO_2

↓

PUPILS
Size, Equal, Reactive

EFR | EMT | EMT ADVANCE | P | AP | EP

ONGOING EXAM Paediatric (≤ 13 years)

PATIENT HISTORY
Complete S.A.M.P.L.E. History if not already done

↓

LOC AVPU
Pupils -
If altered mental status: Finger-stick Glucose

↓

REPEAT INITIAL ASSESSMENT

↓

VITAL SIGNS
BP, HR, RR, BSL, SpO_2

↓

EXAMINE NECK
Obvious Wounds, Tenderness, Swelling , JVD? Tracheal Deviation?

↓

EXAMINE CHEST
Asymmetry (Paradoxical Motion?), Contusions, Penetrations, TIC

↓

BREATH SOUNDS & HEART TONES
(Present? Equal? (If unequal: Percussion)

↓

EXAMINE ABDOMEN
Contusions, Penetration /evisceration; Tenderness, Rigidity, Distension

↓

RECHECK IDENTIFIED INJURIES

↓

CHECK INTERVENTIONS
ET tube ,IVs, seal on chest wound ,decompression needle ,splints & dressings, impaled objects, position of pregnant patients

EFR | EMT | EMT ADVANCE | P | AP | EP

Secondary Survey Trauma Paediatric (≤ 13 years)

Primary Survey

Generalized or unknown mechanism of injury

Yes → S.A.M.P.L.E. History

No → Examination of obvious injuries (Focused exam) → S.A.M.P.L.E. History

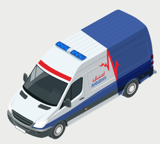
Monitor and record vital signs

GCS

Complete a head to toe survey DCAP-BLS, TIC

Identify positive findings and initiate care management

Go to appropriate CPG



Markers for multi-system trauma
Systolic BP < 90
Respiratory rate < 10 or > 29
Heart rate > 120
AVPU = V, P or U on scale
Mechanism of Injury

Deformities
Contusions
Abrasions
Penetrations
Burns
Lacerations
Swelling
Tenderness
Instability
Crepitus

EFR | EMT | EMT ADVANCE | P | AP | EP

Traumatic Shock – Paediatric (≤ 13years)

INITIAL ASSESSMENT

RAPID TRAUMA SURVEY

Hemorrhagic Shock

- Control bleeding
- Oxygen
- Crystalloid Fluid
10 mL / kg IV/IO
if SBP <70 + 2(age in years) mmHg
- Consider TXA 20 mg/kg slow injection over 10 minutes **(AP,EP)**

Non- Hemorrhagic

- Oxygen
- Identify underlying cause and treat accordingly
- Crystalloid Fluid
10 mL / kg IV/IO
if SBP <70 + 2(age in years) mmHg

Pain Control

Maintain normothermia

Non - Hemorrhagic
Obstructive shock:
- Tension pneumothorax
- Cardiac Tampnade

Neurogenic shock

Shock is defined as impaired tissue perfusion and may be manifested by any of the following:
- Tachycardia (early sign)
- Altered mental status
- Poor skin perfusion
- Low blood pressure

Traditional signs of shock may be absent early in the process.

EFR | EMT | EMT ADVANCE | P | AP | EP

Spinal Immobilisation – Paediatric (≤ 13 years)

NEGATIVE

Mechanism of Injury

POSITIVE

Uncertain

Apply Manual Stabilization Until Exam Complete

Remove helmet (if worn)

Patient in undamaged child seat :
Immobilise in child seat

In Life Threatening :
Rapid extrication with long board/ paediatric board and cervical collar

Positive Mechanisms:

- High-Speed MVC
- Falls > Three Times Patient s Height
- Axial load
- Diving Accidents
- Penetrating Wound In or Near Spinal Column
- Sports Injuries to Head or Neck
- Unconscious Trauma Patient

SPINAL PAIN OR TENDERNESS?

yes

No

MOTOR AND SENSORY EXAM

ABNORMAL

NORMAL

RELIABLE PATIENT?

yes

No

Reliable Patient Is:

- Calm
- Cooperative
- Sober Alert
- Without Distracting Injuries

Unreliable Patient Has:

- Acute Stress Reaction
- Head/Brain Injury
- Altered Mental Status
- Intoxication with Drugs/Alcohol
- Other Distracting Injuries

Negative Spinal Injury: SMR NOT INDICATED

Positive Spinal Injury

Note: If any doubt exists...

Spinal Motion Restriction

EFR	EMT	EMT ADVANCE	P	AP	EP

Burns – Paediatric (≤ 13 years)

Stop the Burning Process

Airway Management
Maintain Ventilation
Manage Circulation

Electrical

ECG

Thermal

Fluid Management for 3rd-degree burns :-
Pediatric :- 3 ml LR x kg x % TBSA

Chemical

Rapid removal of the chemical :
Brush it away,
Remove clothess
or **Irrigating** with water and soap 10-15 minuts.

Cover burn with clean dressing

Prevent hypothermia

Pain Control

Burn (All degree) in following Area Should transport to hospital :
Face & Neck
Hand
Flexion point
Pernium

OBSTETRIC EMERGENCIES

SECTION 4

EFR | EMT | EMT ADVANCE | P | AP | EP

Cardiac Arrest in Pregnancy

Continue BLS/ACLS
- High-quality CPR
- Defibrillation when indicated
- Other ACLS interventions (eg, epinephrine)

↓

Assemble maternal cardiac arrest team

↓

Consider etiology of arrest

↓

Perform maternal interventions
- Perform airway management
- Administer 100% O_2, avoid excess ventilation
- Place IV above diaphragm
- If receiving IV magnesium, stop and give calcium chloride or gluconate

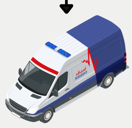
↓

Continue BLS/ACLS
- High-quality CPR
- Defibrillation when indicated
- Other ACLS interventions (eg, epinephrine)

↓

Neonatal team to receive neonate

↓



Maternal Cardiac Arrest

- Team planning should be done in collaboration with the obstetric, neonatal, emergency, anesthesiology, intensive care, and cardiac arrest services.
- Priorities for pregnant women in cardiac arrest should include provision of high-quality CPR and relief of aortocaval compression with lateral uterine displacement.
- The goal of perimortem cesarean delivery is to improve maternal and fetal outcomes.
- Ideally, perform perimortem cesarean delivery in 5 minutes, depending on provider resources and skill sets.

Advanced Airway

- In pregnancy, a difficult airway is common. Use the most experienced provider.
- Provide endotracheal intubation or supraglottic advanced airway.
- Perform waveform capnography or capnometry to confirm and monitor ET tube placement.
- Once advanced airway is in place, give 1 breath every 6 seconds (10 breaths/min) with continuous chest compressions.

Potential Etiology of Maternal Cardiac Arrest

A Anesthetic complications
B Bleeding
C Cardiovascular
D Drugs
E Embolic
F Fever
G General nonobstetric causes of cardiac arrest (H's and T's)
H Hypertension

EFR	EMT	EMT ADVANCE	P	AP	EP

Pre-Hospital Emergency Childbirth

Primary Assessment

- Take OB & SAMPLE history
- Patient in labour/ Imminant Birth — No → transport; Yes ↓
- travel time too long — No → transport; Yes ↓
- Consider ALS
- Position mother
- Monitor vital signs and BP
- Birth Complications — Yes → transport; No ↓
- Support baby throughout delivery
- Dry baby, Stimulate and check ABCs
- Baby stable — No → Go to BLS Neonate CPG; Yes ↓
- Check APGAR Score
- Clamp & cut cord
- Wrap baby using clean sheets
- Mother stable — No → Go to Primary Survey CPG; Yes ↓
- If placenta delivers retain for inspection
- Do not wait on scene for Placenta to deliver transport immediately
- Reassess → transport

Request Maternity Ambulance or Female Responder as required by local policy to come to scene or meet en route

Signs and symptoms of imminent birth
- Regular contractions lasting 45 - 60 seconds at 1- 2 minute intervals
- Mother has urge to bear down or has sensation of bowel movement
- Large amount of bloody show
- Crowning occurs
- Mother believes that delivery is imminent
-Membrane rupture
Gestation < 32 weeks
Cover newborn in poly
thene wrap/bag up to
neck without drying first

Wait at least 1-3 mins post birth, 1st clamp is 6"- 8"(4 fingers), 2nd clamp 2"- 4"(2 fingers), Cut the cord between the 2 clamps

Give the baby to the mother (skin to skin contact) Note time of birth.

- Placenta will deliver spontaneously within 5-15 mins
- Don't force the placenta to deliver
- Don't pull on the umbilical cord
- Contain all tissues in a plastic bag and transport

EFR | EMT | EMT ADVANCE | P | AP | EP

Basic Life Support – Neonate (< 4 weeks)

Antenatal counseling
Team briefing and equipment check

Birth

1 minute

Term gestation?
Good tone?
Breathing or crying?

Yes → **Infant stays with mother for routine care: warm and maintain normal temperature, position airway, clear secretions if needed, dry.**
Ongoing evaluation

No ↓

Warm and maintain normal temperature, position airway, clear secretions if needed, dry, stimulate

Apnea or gasping?
HR below 100/min?

No → **Labored breathing or persistent cyanosis?**

Yes → **Position and clear airway**
SpO_2 monitor
Supplementary O_2 as needed
Consider CPAP

↓ **Postresuscitation care**
Team debriefing

Yes ↓

PPV
SpO_2 monitor
Consider ECG monitor

HR below 100/min?

No → **Postresuscitation care**
Team debriefing

Yes ↓

Check chest movement
Ventilation corrective steps if needed
ETT or laryngeal mask if needed

HR below 60/min?

No → back to **HR below 100/min?**

Yes ↓

Intubate if not already done
Chest compressions
Coordinate with PPV
100% O_2
ECG monitor
Consider emergency UVC

HR below 60/min?

Yes ↓

EMT ADVANCE

IV epinephrine
If HR persistently below 60/min
Consider hypovolemia
Consider pneumothorax

Targeted Preductal SpO_2 After Birth	
1 min	60%-65%
2 min	65%-70%
3 min	70%-75%
4 min	75%-80%
5 min	80%-85%
10 min	85%-95%

EFR | EMT | EMT ADVANCE | P | AP | EP

PV Haemorrhage in Pregnancy

Primary Assessment

Pregnant < 20 weeks
Early pregnancy haemorrhage
→ Apply absorbent pad to perineum area

Pregnancy ≥ 20 weeks
Antepartum haemorrhage
→ Left lateral tilt

Left lateral tilt

↓

Do not examine vagina gently check fetal HR

↓

Apply absorbent pad to perineum area

↓

Oxygen therapy

↓

ECG & SpO_2 & V/S monitoring

↓

Patient is haemodynamically unstable

↓

Crystalloid Fluid IV bolus: 1 - 2 L

↓

Reassess

↓

Contact TeleEMS (severe complaint of pain)

EFR | EMT | EMT ADVANCE | P | AP | EP

Postpartum Haemorrhage

Primary Assessment

After Child birth

Apply absorbent pad to perineum area

Oxygen therapy

Mother is haemodynamically unstable

Yes

Crystalloid Fluid IV bolus: 1 - 2 L

External massage of the uterus

Elevate lower limbs

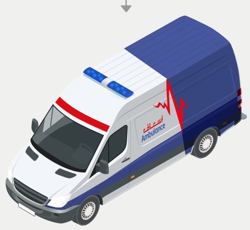
P

Consider

Tranexamic Acid 1 g IV

No

Reassess



Definition of PPH:
Sever vaginal bleeding after Childbirth
Primary: After delivery to 24hours
Secondary : 24hours to 12 weeks

EFR | EMT | EMT ADVANCE | P | AP | EP

Umbilical Cord Complications

Primary Assessment

Request ALS

Request Ambulance Control to contact emergency physician / maternity ambulance /medical team as required by local policy to Come to scene or meet en route.

Oxygen therapy

Cord around baby's neck

Attempt to slip the cord over the baby's head

Successful

Yes

No

Ease the cord from around the neck as shoulders are delivered

Cord rupture

Apply additional clamps to cord

Apply direct pressure with sterile dressing

Go to Childbirth CPG

Prolapsed cord

consider placing mother in prone knee- chest position

Placed gloved hand into vagina and gently lift Head/body off the cord

Asses for pulsation in the cord Maintain until relieved by Hosp Staff.

For prolapsed cord pre-alert hospital as emergency caesarean section will be required

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Breech Birth

Primary Assessment

Request Doctor

Request Ambulance Control to contact emergency physician / maternity ambulance /medical team as required by local policy to Come to scene or meet en route.

No

Consider
Oxygen therapy

Mother to adopt the lithotomy position

Support the baby as it emerges – avoid manipulation of the baby's body

Successful delivery

Yes → Go to Childbirth CPG

No → Support the baby as it emerges – avoid manipulation of the baby's body

No

Nape of neck anteriorly visible at vulva

No → Support the baby as it emerges – avoid manipulation of the baby's body

Yes

Place gloved hand into vagina with fingers between infants face and uterine wall to create an open airway

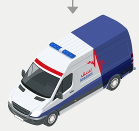
Grasp both baby's ankles in other hand

Rotate baby's legs in an arc in an upward direction as contractions occur

Successful delivery

Yes → Go to Childbirth CPG

No



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Eclampsia (Seizures in Pregnancies)

Eclampsia is pre-eclampsia+Seizure

Pre-eclampsia Symptoms:
- >20 weeks pregnancy
- Hypertension
  SBP>160mmHg
  DBP>110mmHg
- Confusion/AMS
- SOB
- Vaginal bleeding

Primary Assessment
OB History

→ Request ALS

↓

Airway Management
Open Airway+Left Lateral position

↓

High Flow of Oxygen

↓

IV Access

↓

Vital Sign

↓

Magnesium Sulphate 4g IV slowly/ 8mg IM
(4mg in each Gluteal Area)

↓

Still Seizing

No → Reassess

Yes ↓

Benzodiazepines
(Diazepam 5mg IV/ 10mg IM)

↓

Reassess

↓

Pre-Hospital Emergency Care Operations (MCI)

SECTION 5

EFR | EMT | EMT ADVANCE | P | AP | EP

Major Emergency (Major Incident) – First Practitioners on site

Possible Major Emergency

Scene Size Up

Consider pre-arrival information

Responder 1
practitioner who is operating ambulance vehicle

- Park at the scene as safety permits and in conjunction with DCD & Dubai Police if present
- Leave Emergency light on as vehicle acts as Forward Control Point
- Confirm arrival at scene with Ambulance Dispatch and provide an initial visual report stating MCI Standby or Declared
- Maintain communication with Responder 2
- Leave the ignition keys in place and remain with vehicle Carry out

If single Responder is first on site combine both roles until additional Responder arrive

The first ambulance crew does not provide care or transport of patients as this interferes with their ability to liaise with other services, to assess the scene and to provide continuous information as the incident develops

Responder 2
(Ideally higher level practitioner)

Carry out scene survey

Give situation report to Ambulance Dispatch using METHANE message

Carry out Ambulance Incident Commander of Operations **(Ambulance Incident Commander)** role until relieved

Liaise with **(Police Incident Commander)** and **(Fire Incident Commander)**

Select location for **(Ambulance Parking Point)**

Set up key areas in conjunction with other Principal Response Agencies on Scene;

- Scene Control Point (**Ambulance Control Point**),
- Staging area
- Triage Area
- Treatment Area
- Transport Area

METHANE message

M – Major Emergency declaration / standby
E – Exact location of the emergency
T – Type of incident (transport, chemical etc.)
H – Hazards present and potential
A – Access / egress routes
N – Number of casualties (injured or dead)
E – Emergency services present and required

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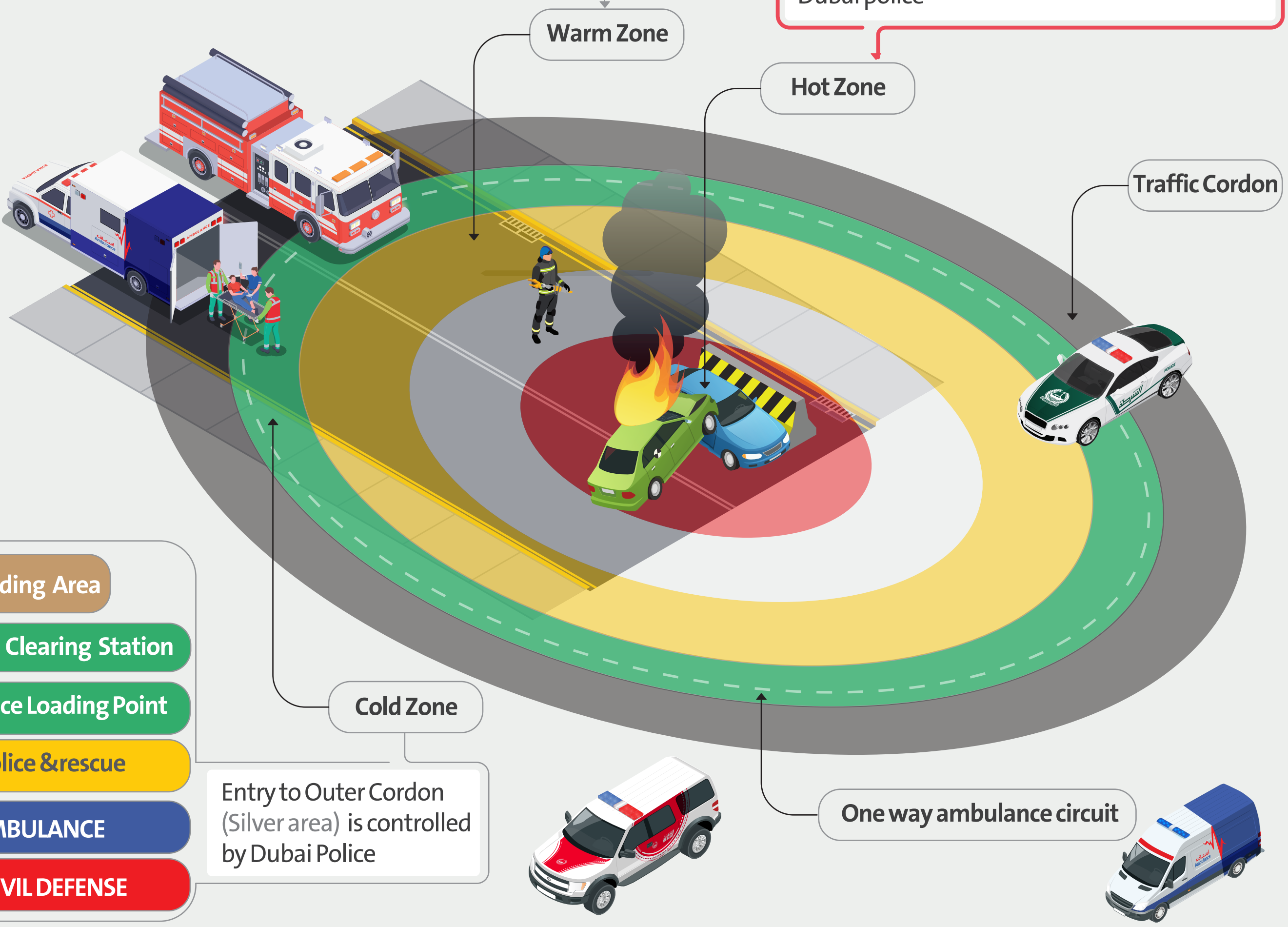
Major Emergency (Major Incident) – Operational Control

Entry to Inner Cordon (Bronze Area) is limited to personnel providing emergency care and or rescue Personal Protective Equipment required

If Danger Area identified, entry to Danger Area is controlled by a Senior Fire Officer or an Dubai police



- Body Holding Area
- Casualty Clearing Station
- Ambulance Loading Point
- Dubai police &rescue
- DCAS AMBULANCE
- DUBAI CIVIL DEFENSE

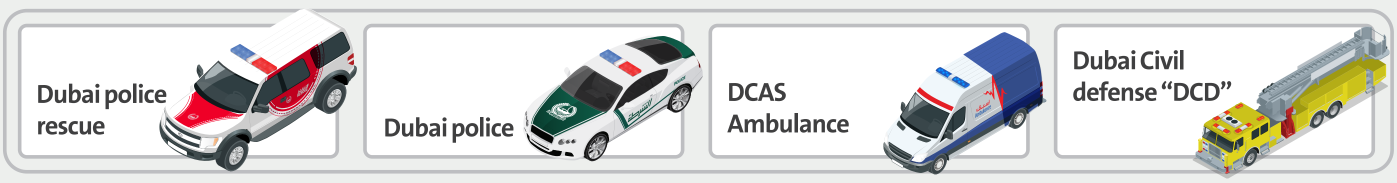
Entry to Outer Cordon (Silver area) is controlled by Dubai Police

Management structure for;
Outer Cordon, Tactical Area (Silver Area)
On-Site Co-ordinator
(Ambulance Incident Commander)
Site Medical Officer **(Medical Incident Commander)**
Local Authority Controller of Operations **(Fire Incident Commander)**
Garda Controller of Operations **(Police Incident Commander)**

Management structure for;
Inner Cordon, Operational Area (Bronze Area)
Forward Ambulance Incident Commander **(Forward Ambulance Incident Commander)** Forward Medical Incident Commander **(Forward Medical Incident Commander)**
Fire Service Incident Commander **(Forward Fire Incident Commander)** Garda Cordon Control Commander **(Forward Police Incident Commander)**

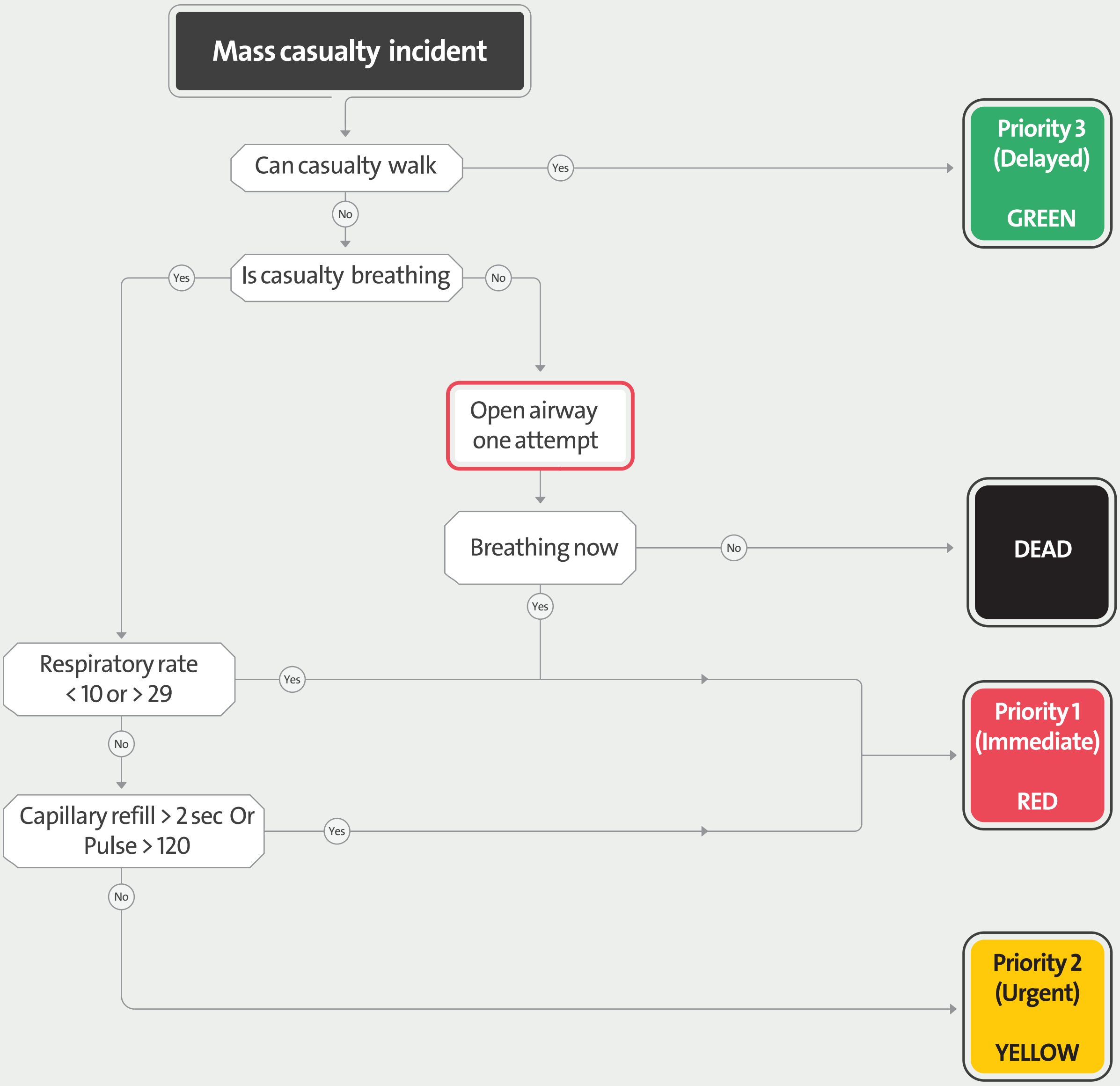
Other management functions for;
Major Emergency site
Casualty Clearing Officer Triage Officer
Ambulance Parking Point Officer Ambulance Loading Point Officer Communications Officer
Safety Officer

Please note that Controller of Operations may be other than ambulance or fire officers, depending on the nature of the emergency



EFR	EMT	EMT ADVANCE	P	AP	EP

Triage Sieve



Triage is a dynamic process

Triage Sort

Multiple casualty incident

Triage is a dynamic process

Cardiopulmonary function	Measured value	Score	Insert score
Respiratory Rate	10–29 / min	4	A
	>29 / min	3	
	6–9 / min	2	
	1–5 / min	1	
	None	0	
Systolic Blood Pressure	≥90 mm Hg	4	B
	76–89 mm Hg	3	
	50–75 mm Hg	2	
	1–49 mm Hg	1	
	No BP	0	
Glasgow Coma Scale	13–15	4	C
	9–12	3	
	6–8	2	
	4–5	1	
	3	0	
Triage Revised Trauma Score			A+B+C

Eye Opening	Spontaneous	4
	To Voice	3
	To Pain	2
	None	1
Verbal Response	Oriented	5
	Confused	4
	Inappropriate words	3
	Incomprehensible sounds	2
	None	1
Verbal Response	Obeys commands	6
	Localises pain	5
	Withdraw (pain)	4
	Flexion (pain)	3
	Extension (pain)	2
	None	1
Glasgow Coma Scale		

Revised Trauma Score

- 1-10 → Priority 1 (Immediate) RED
- 11 → Priority 2 (Urgent) YELLOW
- 12 → Priority 3 (Delayed) GREEN
- 0 → DEAD

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Clinical Care Pathway Decision – Treated at scene

From relevant CPGs

Non serious or non life threat

Patient responds to intervention(s)

No

Administer specific treatment & provide patient with the opportunity to recover/ respond

Yes

Conduct complete patient assessment Focused history Systematic physical examination

All generic inclusion criteria present

No

Yes

Pratient and carer satisfied with Medic

No

Yes

An adult carer both capable & willing to accept responsibility available

No

Yes

Explain Treatment plan and risks

Yes

Exit Scene

Generic patient inclusion

1.≤ 18 years & ≤ 60 years.
2.Not pregnant.
3.Social support available.
4.Demonstrates capacity and willing to engage.
5.Reliable history.
6.Vital signs within normal range (following care).
7.Compliant with treatment, including own medications.
8.Clinical status of 'Non serious or non life threat (following care).
9.Absence of self-inflicted injury or assault.
10.No observed significant relevant co-morbidity.
11. 1st call for same condition within 30 days.
12. Registered with general practitioner.
If in doubt about generic inclusions the practitioner should transport to ED

If Emergency Physician is present; follow direction on transport decision

Treatment plan and risks pathway:

1. Explain treatment plan to the patient and carer.
2. Explain risks and benefits of treatment.
3. If patient insists on going to the hospital then transport patient to the nearest hospital

EFR | EMT | EMT ADVANCE | P | AP | EP

Handling the Dead on Scene

Verification of death

CIRCULATORY
No pulse present No heart sounds Asystole on ECG

RESPIRATORY
No respiratory effort No chest sounds

NEUROLOGICAL
Pupils not responding to light No reaction to painful stimuli

An unexpected death has occurred
An expected death has occurred
Cease resuscitation has occurred
An obvious dead body has been found

It is not uncommon for there to be occasional spontaneous gasping sounds or occasional intermittent audible heart sounds soon after an expected death and if this is the case the patient should be left for 15 minutes and the full procedure repeated.

Verify death using criteria outlined

Inform Ambulance Dispatch

Ambulance Dispatch to inform police

Record Death in Patient Records(EPCR)

Complete Verification of Death Record Form

Provide initial support to family/carers

Await arrival of Police Patrol
(unless responding to a Life threatening call as adviced by Dispatch)

In case of suspected criminal death await forensics team to arrive on scene for additional information.

Police on Scene

No

Yes

Exit the scene

MEDICATION FORMULARY

SECTION 8

Medication	ADENOSINE
Authorisation	P AP EP
Classification	Antiarrhythmic
Presentation	Vial containing 6 mg adenosine in 2 ml
Indications	• Conversion of Paroxysmal Supraventricular Tachycardias
Administration and Dosages	**Adult:** • Initial dose is a 6 mg rapid bolus over 2 seconds into large proximal peripheral vein followed by a rapid 20 ml flush of normal saline • If no response within 2–1 minutes, a 12 mg IV/IO repeat dose should be administered – Patients with a heart transplant are very sensitive to effects of adenosine and should receive initial dose of 3mg over 2 seconds, followed if necessary by 6 mg after 2–1 minutes. **Paediatrics:** • Initial dose 0,1 mg/kg (to a maximum of 6 mg) followed by a rapid 10 ml flush of saline. • If no response within 2–1 minutes, a repeat 0,2 mg/kg (to a maximum of 12 mg) dose should be administered. • Patients with a heart transplant are very sensitive to effects of adenosine and should receive initial dose of 3 mg over 2 seconds, followed if necessary by 6 mg after 2–1 minutes
Actions	Slows conduction through the AV node, interrupts AV-nodal re-entry pathways and can restore normal sinus rhythm in PSVT. ONSET: immediate, PEAK: approx. 20 seconds DURATION: approx. 40 seconds.
Contraindication	• Convulsions, Dizziness, Headache, Weakness • Palpitations, Sweating, Syncope, Feeling of impending doom • Bronchospasm, Hyperventilation, Dyspnoea • Nausea/Vomiting, Metallic taste, Blurred Vision, Flushing • Injection-site reactions • Cardiac Arrest • Worsening of intracranial hypertension .
Side effects	• Convulsions, Dizziness, Headache, Weakness • Palpitations, Sweating, Syncope, Feeling of impending doom • Bronchospasm, Hyperventilation, Dyspnoea • Nausea/Vomiting, Metallic taste, Blurred Vision, Flushing • Injection-site reactions • Cardiac Arrest • Worsening of intracranial hypertension
Additional information	• When clinically advisable, use appropriate vagal manoeuvres prior to adenosine administration • At the time of conversion to normal sinus rhythm, a variety of new rhythms may appear on ECG including: short period of asystole, premature ventricular contractions, atrial premature contractions, sinus bradycardia, sinus tachycardia, skipped beats, AV nodal block

Medication	AMIODARONE
Authorisation	EMT ADVANCE, P, AP, EP
Classification	Antiarrhythmic
Presentation	Please remember drug presentations change! Check before administration! • Vial 150 mg in 3 ml
Indications	Please remember drug presentations change! Check before administration! • Vial 150 mg in 3 ml
Administration and Dosages	**Adult: > 18 yrs:** • Cardiac Arrest IV/IO: 300mg rapid push initial dose, followed (once) by 150mg IV/IO push if still in VT/VF. Administer refractory to the 3rd and 5th shock • Tachycardia IV: 5mg/kg in 250mls dextrose %5 over 120 – 20 min **Paediatric: < 18 yrs:** • 1 month – 18 yrs IV/IO: 5 mg/kg bolus over 3 minutes (Maximum of 300 mg), repeat once. Administer refractory to the 3rd and 5th shock • Neonate IV/IO: 5 mg/kg bolus over 3 minutes, repeat once. Administer refractory to the 3rd and 5th shock
Actions	• Class 3 antiarrhythmic, lengthens cardiac action potential and therefore effective refractory period and QT interval on ECG • Prolongs atrioventricular conduction • Blocks potassium channels in cardiac muscle • Significant sodium channel blocking activity • Acts to stabilise and reduce electrical irritability of cardiac muscle.
Contraindication	• No contraindications in the context of the treatment of cardiac arrest
Caution	• Patients already receiving beta and calcium channel blockers and anti-arrhythmic medication, however in cardiac arrest amiodarone should not be withheld
Side effects	• Bradycardia • Torsades de pointes • Vasodilation causing hypotension flushing • Bronchospasm • Injection-site reactions • Cardiac Arrest • Worsening of intracranial hypertension
Additional information	• Administer into large vein as extravasation can cause irritation/necrosis • Never to be given via endotracheal route • Should be followed by 0.9% NaCl flush • Draw up slowly to prevent bubbling • For ease of use in the paediatric calculations when using 150mg in 3 ml ampoule, add 2mls Dextrose 5% making a concentration of 150mg in 5ml. • ECG monitoring and resuscitation facilities must be available during IV use

Medication	ASPIRIN
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Platelet aggregator inhibitor
Presentation	81mg chewable tablet
Indications	• Ischaemic myocardial chest pain/suspected myocardial infarction
Administration and Dosages	324 mg ONCE
Actions	• Anti-inflammatory agent and an inhibitor of platelet function.
Contraindication	• Paediatric: Contraindicated
Caution	• Asthma • Pregnancy • Kidney or liver failure • Gastric or duodenal ulcer • Current treatment with anticoagulants
Side effects	• Gastric bleeding • Epigastric pain and discomfort • Wheezing in some asthmatics
Additional information	• Aspirin is contra-indicated in children under the age of 16 years as it may precipitate Reye's syndrome. • In suspected MI, administer 324 mg aspirin regardless of any previously taken that day

Medication	ATROPINE
Authorisation	P AP EP
Classification	Anticholinergic/Vagolytic
Presentation	Prefilled syringe containing 0.5 mg atropine sulphate in 5 ml
Indications	• Unstable bradycardia • Organophosphate poisoning
Administration and Dosages	**Adult:** • Symptomatic Bradycardia IV/IO: 0,5 mg push, repeat at 5–3 min intervals to max 3 mg • Organophosphate poisoning IV/IO: 1 mg, repeat at 5–3 minute intervals until muscarinic symptoms reverse **Paediatrics:** • Symptomatic Bradycardia IV/IO: 0,02 mg/kg push, repeat ONCE. Single dose (max 0,5mg) • Organophosphate poisoning IV/IO: initial dose 0,05–0,02 mg/kg bolus, repeat every 30–20 minutes until muscarinic symptoms reverse
Actions	• May reverse effects of vagal overdrive • May increase heart rate by blocking vagal activity in sinus bradycardia, second- or third-degree heart block • Enhances SA node automaticity • Enhances A-V conduction • ONSET: immediate PEAK: 2–4 minutes DURATION: 4 hours
Contraindication	• Hypothermic bradycardia
Caution	• May induce tachycardia when used after myocardial infarction, which will increase myocardial oxygen demand and worsen ischaemia. Hence, bradycardia in a patient with an MI should ONLY be treated if the low heart rate is causing problems with perfusion such as hypotension
Side effects	• Tachycardia • Dry mouth • Dilated pupils and visual blurring • Confusion and hallucination
Additional information	• Hypoxia is the most common cause of bradycardia in children, therefore interventions to support ABC's, and oxygen therapy should be the first-line therapy. • Symptoms of poisoning reversal include skin flushing, pupil dilation, and bradycardia resolves

Medication	CLOPIDOGREL
Authorisation	EMT ADVANCE, P, AP, EP
Classification	Antiplatelet
Presentation	Please remember drug presentations change! Check before administration! 300 mg tablet or 75mg tablet
Indications	• Acute ST segment elevation myocardial infarction (STEMI) in patients: • Receiving thrombolytic treatment • Anticipated thrombolytic treatment • Anticipated percutaneous coronary intervention (PPCI) Not already taking clopidogrel
Administration and Dosages	**Adult:** > 18 – 75 yrs: PO: • Anticipated thrombolysis: 300 mg • Anticipated PPCI: 600 mg* *Note: If a patient is anticipated to receive PPCI and has already taken 300 mg, an additional 300 mg can be administered (to take total dose to 600 mg) > 75 yrs: PO: • 75 mg regardless of anticipated procedure type **Paediatric:** Not indicated
Actions	Clopidogrel selectively inhibits the binding of adenosine diphosphate (ADP) to its platelet receptor, and the subsequent ADP-mediated activation of the GPIIb/IIIa complex, thereby inhibiting platelet aggregation. Biotransformation of clopidogrel is necessary to produce inhibition of platelet aggregation. Clopidogrel acts by irreversibly modifying the platelet ADP receptor ONSET: 2 hrs PEAK: 1 hr DURATION: 5 days
Contraindication	• Known allergy or sensitivity to clopidogrel • Severe liver impairment • Active pathological bleeding such as peptic ulcer or intracranial haemorrhage • Breastfeeding CAUTIONS: As the likely benefits of a single dose of clopidogrel outweigh the potential risks, clopidogrel may be administered in: • Pregnancy • Patients taking non-steroidal anti-inflammatory drugs (NSAIDs) • Patients with renal impairment
Side effects	• Dyspepsia • Abdominal pain • Diarrhoea • Bleeding (gastrointestinal and intracranial) – the occurrence of severe bleeding is similar to that observed with the administration of aspirin
Additional information	• To be administered in conjunction with aspirin unless there is a known contraindication to aspirin therapy

Medication	COMBIVENT
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Combination of Ipratropium Bronchodilators and Salbutamol ß 2 Agonist
Presentation	Nebules containing ipratropium bromide 0.5mg, salbutamol 3mg in 2.5ml.
Indications	• Asthma • COPD • Severe allergic reaction • Pulmonary edema • Smoke inhalation • Exercise induced bronchospasm • Bronchospasm unresponsive to salbutamol
Administration and Dosages Actions	**Adult:** • 2,5ml, add to 2ml sterile water **Paediatric:** • Not recommended for children under 12 years.
	• Inhibits interaction of acetylcholine at receptor sites on bronchial muscle causes relaxation of bronchial muscle that result in bronchodilation.
Contraindication	• Hypersensitivity to Atropine • Hypersensitivity to soybean products • Hypersensitivity to peanuts • Hypertrophic obstructive cardio- myopathy or tachyarrhythmia
Caution	• May induce tachycardia when used after myocardial infarction, which will increase myocardial oxygen demand and worsen ischaemia. • Hence, bradycardia in a patient with an MI should ONLY be treated if the low heart rate is causing problems with perfusion such as hypotension
Side effects	• Palpitation, • Dizziness, • Anxiety • Headache • Muscle cramp • Dry mouth

Medication	DEXAMETHASON
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Corticosteroid, glucocorticoid
Presentation	Bottle containing 0.5mg/5ml syrup 100ml.
Indications	• Croup (mild. Moderate, sever) • Bronchospasm • Asthma. • Anaphylaxis.
Administration and Dosages	• PO: 0.15mg/kg - 0.6mg/kg (Max. dose 16mg)
Actions	• Anti-inflammatory or Immunosuppressive agent. Supress allergic disorder
Contraindication	• Known hypersensitivity • Cushing's syndrome • Systemic Fungal Infection • Cerebral Malaria
Caution	None
Side effects	• Hyperglycaemia • GI discomfort • Immunosuppression
Additional information	• Administer with juice if available • Discard after single use

Medication	DEXTROSE 10%
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Antihypoglycaemic Agent
Presentation	500 ml of 10% glucose solution (50 g)
Indications	• Hypoglycaemia (blood glucose < 4.0 mmol/l OR 72 mg/ dl) • Clinically suspected hypoglycaemia where oral glucose administration is not possible • Unconsciousness where likely cause is hypoglycaemia • In Cardiac Arrest: Hyperkalaemia (> 6.5 mmol) (ESP only)
Administration and Dosages	**Adult:** > 12 yrs: • IV/IO: 10 g (100 ml) repeat every 5 minutes. Maximum dose 30 g • Titrate to serum glucose level as per protocol • If no response after 2nd dose rapidly transport to hospital and pre-alert. Consider differential diagnosis **Paediatric:** < 12 yrs: • IV/IO: 1 month to 12 yrs: 0.5 g/kg repeat every 5 minutes. Maximum of 3 doses • IV/IO: Neonate: 0.9 g repeat every 5 minutes. Maximum of 3 doses • Titrate to serum glucose level as per protocol • If no response after 2nd dose rapidly transport to hospital and pre-alert. Consider differential diagnosis
Actions	• Reversal of hypoglycaemia • Glucose is a monosaccharide or simple sugar or carbohydrate ONSET: immediate PEAK: 15 minutes
Contraindication	None
Caution	• Administer by large bore cannula into a large vein to avoid extravasation. Concentration of %10 glucose is an irritant to veins • Use caution when administering intravenous dextrose solutions to patients with renal impairment and particularly neonates, low birth weight infants, and premature neonates since dextrose solution contains aluminum, which may be toxic
Side effects	• Hyperglycaemia • Electrolyte disturbance
Additional information	• Use all clinical information available so you are able to make an informed decision between glucose 10% IV, glucose gel 40% PO, or glucagon IM • Glucose and dextrose administration pose no particular risks during breastfeeding • Electrolyte imbalance, particularly in serum potassium and phosphate, may occur during prolonged use of concentrated dextrose solutions. The intravenous administration of dextrose solutions can cause fluid and/or solute overload resulting in dilution of serum electrolyte concentrations, overhydration, congested states (congestive heart failure) or pulmonary oedema • Clinically, no drug interactions are of concern as long as blood glucose and other serum chemistries are monitored • Allergic reactions after administration of dextrose injection are very rare.

Medication	DIAZEPAM
Authorisation	EP
Classification	Benzodiazepines
Presentation	• 10MG/2ML INJECTION Solution for injection or infusion
Indications	• Status epilepticus • Convulsions • Agitation.
Administration and Dosages	**Adult:** • 5 to 10 mg IV / IM once, repeated at 10-to-15-minute intervals to a maximum dose of 30 mg if necessary **Paediatrics:** • 0.5mg/kg rectal (max 10mg) 0.2mg/ kg IV/IM (max 10mg)
Actions	• Diazepam acts on central nervous system by stimulating GABA (gamma amino butyric acid) which inhibits the excitatory stimulation in the brain. It suppresses the activity of brain and suppresses the convulsions. It potentiates the inhibitory neurotransmitters in the limbic system and there by decreases the emotions and anxiety.
Contraindication	• Respiratory depression
Caution	Diazepam has a risk for abuse and addiction, which can lead to overdose and death
Side effects	• Affect ability to drive and use machines
Additional information	• Diazepam has a risk for abuse and addiction, which can lead to overdose and death • Narcotics and controlled medications policy applicable

Medication	DICLOFENAC
Authorisation	EMT ADVANCE, P, AP, EP
Classification	Nonsteroidal Ant-Inflammatory
Presentation	75mg/1ml ampoule
Indications	• Mild to moderate pain and inflammation. • Rheumatic disease, musculoskeletal disorders and acute gout.
Administration and Dosages	**Adult: > 16yrs:** • PO: 50 mg once every 8 to 12 hours. Maximum cumulative dose 150 mg in 24hrs • Should not be administered if already administered in the last 8hours **Paediatric: < 16yrs:** • not indicated
Actions	• Reduces the production of prostaglandins by inhibiting the enzyme cyclooxygenase, thus reducing inflammation and pain ONSET: 30 mins – 1hr
Contraindication	• Known hypersensitivity to aspirin or any other NSAID < 16 Years • Clotting disorders (e.g. Haemophilia, Christmas disease & Von Willebrand disease) • Ischaemic heart disease • Cerebrovascular disease • Peripheral arterial disease • Active gastrointestinal ulceration or bleeding • Pregnancy • Significant renal impairment • Should not be administered if already administered in the last 8 hours
Caution	• Hypertension • Crohn's disease or ulcerative colitis • Paracetamol is preferred for use in the elderly • Heart failure
Side effects	• Nausea and Diarrhoea • Gastric bleeding and ulceration • Headache • Nervousness • Drowsiness • Vertigo, Dizziness, Tinnitus • Photosensitivity • Hypersensitivity reactions • Depression • Insomnia • Haematuria
Additional information	• This drug is becoming less common and being replaced with more desirable drugs with less side effects. • May cause an increased risk of serious cardiovascular thrombotic events, myocardial infarction, and/or stroke, • which can be fatal. Patients with cardiovascular disease or risk factors for cardiovascular disease may be at greater risk.

Medication	DIPHENHYDRAMINE
Authorisation	P AP EP
Classification	Antihistamine /H1 blocker
Presentation	Ampoule: 50mg /1mL
Indications	• Allergic Reaction (urticarial and/or pruritis) • Dystonia/akathisia
Administration and Dosages	**Adult :** **Anaphylaxis/Allergic Reaction** • IV/IM: 50mg **Dystonia Reaction:** • IV/IO/IM: 50-25mg **Paediatric:** **Anaphylaxis/Allergic Reaction** • IV/IO/IM: 1 MG/KG (MAX 50mg) **Dystonia Reaction:** • IV/IM: 2-1mg/kg
Actions	• Block the cellular histamine receptors resulting in decreased capillary permeability, decreased itching, edema, bronchoconstriction, and vasodilation
Contraindication	• Hypersensitivity • premature infants and neonates
Caution	None
Side effects	• Drowsiness. • Sedation. • Dry mouth and throat. • CNS depression. • Cardiac arrhythmia (sodium channel blockade) • Seizure.
Additional information	None

Medication	DOPAMINE
Authorisation	EP
Classification	Sympathomimetic, Inotrope, Vasopressor
Presentation	Ampoule containing : 200mg/5ml (40mg/ml) - 400mg/10ml (40mg/ml)
Indications	• Hypotension caused by: • Septic shock • Bradycardia • Myocardial infarction • Trauma/spinal shock • Heart Failure (cardiogenic shock)
Administration and Dosages	**Adult:** Shock: • IV: 50-1mcg/kg/min (MAX 5-20-mcg/kg/min) Post-resuscitation stabilization: • IV: 10-5mcg/kg Cardiogenic shock due to heart failure: • IV: 3-1mcg/kg/min Bradycardia: (if patient not responsive to Atropine): • IV: 10-2mcg/k/min Dopamine infusion: • Mix 400mg dopamine into 250ml NS/D5W, using microdrop (60gtt/ml) set Make 1,6mg/ml **NOTE: if heart rate exceeds 140bpm-the infusion should be discontinued. can cause hypertension crisis** **Paediatric Dosing:** Shock: 5 • 10mcg/kg/min (MAX 20mcg/kg/min) Post-resuscitation stabilization: IV/IO: 2- • 20mcg/kg/min
Actions	• Block the cellular histamine receptors resulting in decreased capillary permeability, decreased itching, edema, bronchoconstriction, and vasodilation
Contraindication	• Hypersensitivity • Tachyarrhythmia • Pheochromocytoma • Hypovolemia due to trauma
Caution	None
Side effects	• Anaphylaxis. • Hypertension • Bronchospasm • Ventricular Arrhythmia • Atrial Fibrillation • Palpitation • Angina • Dyspnoea • Tachycardia • Headache
Additional information	None

Medication	EPINEPHRINE 1:1,000 (1MG/ML)
Authorisation	**Depends on Case HIGH ALERT MEDICATION**
Classification	Sympathetic Agonist
Presentation	Ampoule containing 1 mg adrenaline in 1 ml.
Indications	• Paramedic and above: • Anaphylaxis • Reversal of laryngeal oedema & bronchospasm • Life threatening asthma • Croup/Epiglottitis • Advance Paramedic and Emergency Physician: • Shock: including Sepsis • Unstable bradycardia unresponsive to TCP and Atropine • Post cardiac arrest care unresponsive to fluid therapy • All Levels: • Cardiac arrest
Administration and Dosages	• Adult: • IM Anaphylaxis, laryngeal oedema & bronchospasm, life threatening asthma: 0.5 mg (0.5 ml), repeat every 5 minutes until symptoms relieved • IV/IO cardiac arrest: dilute epinephrine 1:1000 1mg/1ml ampule with 9ml of Normal saline to create 1:10000 1mg/10ml. Administer 1mg , repeat every 3-5mins • IV/ IO Drip: Shock, Unstable bradycardia & Post cardiac arrest care: 1 mg mixed in 500 ml normal saline (concentration of 2 mcg/ml). Administer 2 mcg/min (1ml/min) Maximum rate 10 mcg/min • IV Infusion: 2 – 10 mcg/min • Paediatric: Refer to Broselow Tape
Actions	• Reverses allergic manifestations of acute anaphylaxis • Relieves bronchospasm in acute severe asthma • ONSET: 3 to 10 minutes, IM PEAK: 20 minutes, IM DURATION: 20 to 30 minutes
Contraindication	None
Caution	• Severe hypertension may occur in patients on betablockers, and half doses should be administered unless there is profound hypotension • For patients taking tricyclic anti-depressants (e.g., amitriptyline, imipramine) half doses of adrenaline should be administered for anaphylaxis Do not give repeated doses of adrenaline to hypothermic patients
Side effects	• Palpitations • Respiratory difficulty • Restlessness • Hypertension • Angina like symptoms
Additional information	• Protect vials from light • Do not use solution if brown in colour or contains precipitate

Medication	FUROSEMIDE
Authorisation	P AP EP
Classification	Loop diuretic
Presentation	Ampoule containing 20 mg furosemide in 2 ml
Indications	• Adjunct in treatment of acute pulmonary oedema secondary to left ventricular failure
Administration and Dosages	**Adult: > 18 yrs:** • IV/IO: 40 mg slow injection over 2 minutes. ONCE **Paediatric: < 18 yrs:** • Not indicated
Actions	• A non K+ sparing potent loop diuretic that inhibits sodium and chloride re-absorption at the proximal and distal tubules and ascending Loop of Henle • A potent diuretic with rapid onset and half life ONSET: 5 minutes PEAK: 30 minutes DURATION: 2 hours
Contraindication	• Hypersensitivity to the medication • Severe renal failure with anuria • Pre-comatose state secondary to liver cirrhosis • < 18 yrs
Caution	• Hypokalaemia could induce arrhythmias • Pregnancy • Hypotension – avoid administration if patients systolic BP 90 mmHg or less
Side effects	• Electrolyte imbalance (particularly hypokalaemia) • Volume depletion • Postural hypotension • GI disturbances • Deafness or tinnitus with large parenteral doses • Dermatitis, pruritus, blurred vision, dizziness, bladder spasms, pancreatitis, thrombocytopenia, agranulocytosis • Very rarely cardiac arrest has occurred after IV or IM administration (may occur especially in digitalised patients with hypokalaemia)
Additional information	• Nitrates are the first line treatment for acute pulmonary oedema. Use furosemide as secondary treatment where transfer times are prolonged • Co-administration with catecholamines (i.e. dopamine, dobutamine, epinephrine): precipitation islikely to occur (avoid simultaneous infusion/injection via same IV line). • Many incompatibilities; do not mix with any other injectable drugs • Monitor BP and pulse rate routinely • Monitor blood glucose in diabetic patients • Watch for signs of hypokalaemia • May precipitate gout; hyperglycaemia in diabetics; vascular thrombosis in elderly or debilitated

Medication	GLUCAGON
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Anti-hypoglycaemic
Presentation	1 mg glucagon powder for reconstitution with water for injection
Indications	• Hypoglycaemia (blood glucose < 4.0 mmol/l or 72 mg/dl) especially in known diabetics • Clinically suspected hypoglycaemia WHERE ORAL GLUCOSE ADMINISTRATION IS NOT POSSIBLE • The unconscious patient, where hypoglycaemia is considered a likely cause • Calcium channel blocker & Beta blocker overdose (Paramedic and above only)
Administration and Dosages	• Adult: • IM: Hypoglycaemia: 1 mg ONCE only • IV: Calcium channel blocker & Beta blocker overdose: 2 mg bolus ONCE only (Paramedic and above only). • Paediatric: • IM: Hypoglycaemia: – 1 month – < 8 years: 0.5 mg ONCE • Neonate IM: 0.1 mg ONCE • IV: Calcium channel blocker & Beta blocker. • overdose: NOT INDICATED
Actions	• Glucagon is a hormone that induces the conversion of glycogen to glucose in the liver, thereby raising blood glucose levels • ONSET: 4 – 10 minutes • PEAK: No documentation found DURATION: 10 – 30 minutes
Contraindication	• Low glycogen stores (e.g. alcoholics, malnourished, cachexic patients, recent use of glucagon) • Known severe adverse reaction • Phaeochromocytoma
Caution	• Avoid intramuscular administration of any drug when a patient islikely to require thrombolysis • Use caution when administering in active seizures caused by hypoglycaemia, needlestick injuries may occur, follow sharps protocol
Side effects	• Nausea • Vomiting • Diarrhoea • Acute hypersensitivity reaction (rare) • Hypokalaemia • Hypotension • Bronchospasm
Additional information	• Where ALS is not present BEFORE administering Glucagon, contact CSD to confirm appropriateness • Oral carbohydrates, such as bread and bananas, are the preferred method of treatment where possible • Give complex carbohydrates as mentioned above after administration of glucagon to replenish glycogen levels • Confirm effectiveness by checking blood glucose 5 – 10 minutes after administration • Glucagon may be required where IV Glucose is unavailable and the patients responsiveness is P/U on AVPU assessment. Oral carbohydrate/oral glucose may be more appropriate where response is A/V on AVPU assessment • Glucagon is relatively ineffective once body glycogen stores have been exhausted (especially hypoglycaemic, non-diabetic children). In such patients, use oral glucose gel smeared round the mouth/gums or glucose 10% IV as first-line treatments • The neonate's liver has very limited glycogen stores, so glucagon may not be effective • Glucagon may also be ineffective in some instances of alcohol-induced hypoglycaemia

Medication	GLUCOSE 40% ORAL GEL
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Oral anti-hypoglycaemic
Presentation	Plastic tube containing 25 g glucose 40% oral gel
Indications	• Known or suspected hypoglycaemia (< 4 mmol/l or 72 mg/dl) in a conscious patient where there is no risk of choking or aspiration
Administration and Dosages	**Adult: > 12 yrs:** • One-half tube – buccal administration. Repeat every 5 minutes as required **Paediatric: < 12 yrs:** • One-half tube – buccal administration. An appropriate amount should be given considering the child's age. Repeat every 5 minutes as required
Actions	Rapid increase in blood glucose levels via buccal absorption
Contraindication	• Unconscious patient • Known severe adverse reaction
Caution	• Altered level of consciousness where there is risk of choking or aspiration
Side effects	• May cause vomiting in patients under the age of five if administered too quickly
Additional information	• Can be repeated as necessary in hypoglycaemic patient • Treatment failure should prompt the use of an alternative such as glucagon IM or glucose 10% IV • Where there is risk of choking or aspiration glucose gel may be administered by soaking a gauze swab and placing it between the patient's lips and gum to aid absorption. USE EXTREME CAUTION • Treatment should be used in conjunction with blood glucose checks every 5 minutes • After a full assessment and improvement in blood glucose levels ensure the patient eats complex carbohydrates such as bread and bananas

Medication	GTN SL (SUBLINGUAL)
Authorisation	EMT ADVANCE · P · AP · EP
Classification	Nitrate, anti-anginal
Presentation	400 micrograms per metered dose, sublingual spray
Indications	• Cardiac chest pain due to angina or myocardial infarction • Acute pulmonary oedema
Administration and Dosages	**Adult:** • 0.4 mg (400 mcg). Administer 1 spray sublingual. Repeat at 3 – 5 min intervals prn **Paediatric:** • not indicated (unless already prescribed)
Actions	Releases nitric oxide which acts as a vasodilator, resulting in: Dilatation of coronary arteries/relief of coronary spasm Dilatation of systemic veins resulting in lower preload Reduced blood pressure ONSET: < 2 minutes PEAK: 4 – 10 minutes DURATION: 10 – 30 minutes
Contraindication	• Hypersensitivity to the medication • Hypotension (systolic blood pressure < 90mmHg) • Hypovolaemia • Head trauma • Cerebral haemorrhage • GTN must not be given to patients who have taken any PDE5 inhibitors (Sildenafil (Viagra), Tadalafil (Cialis) or Vardenafil (Levitra)) within the previous 24 hours. Profound, refractory hypotension may occur • Unconscious patients Right ventricular infarction
Caution	None
Side effects	• Dizziness • Transient Hypotension • Flushing • Headache
Additional information	• Hold the pump spray vertically with the valve head uppermost • Place as close to the mouth as possible and spray under the tongue • The mouth should be closed after each dose • Failure to deliver a SL administration (i.e. spraying on top of the tongue instead) will result in 90% of the drug being hepatically metabolised before reaching the systemic circulation • If the pump is new or it has not been used for a week or more the first spray should be released into the air to prime the pump

Medication	HALOPRIDOL
Authorisation	EP
Classification	Antipsychotic
Presentation	Ampoule containing 5 mg haloperidol in 1 ml
Indications	• For the management of acute psychosis or agitated/violent behaviour refractory to non-pharmacologic interventions
Administration and Dosages	**Adult: IM:** • Acutely disturbed: 0,2 – 0,1 mg/kg • IM: 10 – 5 mg Repeat in 20 minutes (Maximum cumulative dose 10 mg) **Paediatric: < 12 :** • not recommended
Actions	Haloperidol blocks postsynaptic dopamine D1 and D2 receptors in the mesolimbic system and decreases the release of hypothalamic and hypophyseal hormones. It produces calmness and reduces aggressiveness with disappearance of hallucinations and delusions. ONSET: 30 – 60 mins
Contraindication	**< 12 years** • Movement disorder (e.g. Parkinson’s disease, Tardive dyskinesia) • Severe liver disease • Cardiovascular disease and elongated QT-interval • Bradycardia • CNS depression • IV administration in combative patient
Caution	• Epilepsy • Renal impairment (use smaller doses) • Neonatal thrombocytopenia reported, but risk should be balanced against risk of uncontrolled maternal hypertension; manufacturer advises avoid before third trimester • Elderly patients
Side effects	• Hypersensitive reaction • Dyspnoea • Oedema • Rarely bronchospasm • Hypoglycaemia • Toxic epidermal necrolysis (TEN) • Hypertension • Sweating • Stevens-Johnson syndrome (SJS) • Inappropriate antidiuretic hormone secretion
Additional information	• Protect from light (slight yellowing of injection is common) • Can be mixed in the same syringe as lorazepam or midazolam • Administration to < 18 yrs efficacy unknown • Smaller doses should be used in elderly patients (2.5 – 5 mg) • Administer haloperidol refractory to nonpharmacological attempts at de-escalation • Lost or stolen Haloperidol should be reported immediately to pharmacist in-charge, Medical Director and duty shift in-charge.

Medication	HYDROCORTISONE
Authorisation	EMT ADVANCE, P, AP, EP
Classification	Sympathetic glucocorticoid
Presentation	100 mg Vial
Indications	• Severe or recurrent anaphylactic reactions • Asthma refractory to Salbutamol and Ipratropium Bromide • Exacerbation of COPD • Adrenal insufficiency
Administration and Dosages	**Adult: > 12 yrs:** • IV/IM: Anaphylactic reaction/Exacerbation of COPD: 200 mg slow injection over 2 minutes • IV/IM: Asthma and Adrenal insufficiency: 100 mg slow injection over 2 minutes **Paediatric: < 12 yrs:** • IV/IM: Anaphylactic reaction and Asthma: • < 1 yr: 25 mg slow injection over 2 minutes • 1 to 5 yrs: 50 mg slow injection over 2 minutes • 5–12 yrs: 100 mg slow injection over 2 minutes IV/IO/IM: Adrenal insufficiency: • 6 months to 5 yrs: 50 mg slow injection over 2 minutes • 12–5 yrs: 100 mg slow injection over 2 minutes
Actions	• A synthetic glucocorticoid used as an anti-inflammatory or immunosuppressive agent • Potent anti-inflammatory properties and inhibits many substances that cause inflammation
Contraindication	• No major contraindications in acute management of anaphylaxis
Caution	• None relevant to a single dose. • Avoid intramuscular administration if patient likely to require thrombolysis
Side effects	• CCF • Hypertension • abdominal distension • vertigo • headache • nausea • malaise • hiccups
Additional information	• Intramuscular injection should avoid the deltoid area because of the possibility of tissue atrophy • Dosage should not be less than 25 mg • IV is the preferred route for adrenal crisis • Intravenous SLOW injection over a minimum of 2 minutes to avoid side effects

Medication	HYOSCINE BUTYLBROMIDE
Authorisation	P AP EP
Classification	Anticholinergic /Antispasmodic
Presentation	Ampoule : 20mg /ml
Indications	• Symptomatic relief of gastro-intestinal or Genito-urinary disorder like: • Abdominal Cramps due to gastroenteritis • Stomach pain • Renal Colic • Dysmenorrhea • Irritable bowel diseases
Administration and Dosages	**Adult:** • IM/IV/SC: 20mg **Paediatric:** • Not recommended
Actions	Smooth muscle relaxing /spasmolytic effect
Contraindication	• Paralytic ileus • Pyloric stenosis • Prostatic enlargement • Myasthenia gravis • Renal Impairment. • Constipation
Caution	None
Side effects	• Photophobia • Transient bradycardia (followed by tachycardia, palpitation and arrhythmia). • Reduce d bronchial secretions • Urinary urgency and retention • Dilatation of pupil with loss of accommodation • Dry mouth • Constipation **Occasional side effect:** • Confusion, vomiting, nausea and giddiness • Make sure to take the patient history of substance abuse or addition.
Additional information	None

Medication	KETAMINE
Authorisation	EP
Classification	Dissociative Anaesthetic
Presentation	• Ampoule containing 50 mg/1ml-500mg ketamine in 10 ml
Indications	• Analgesia for moderate to severe pain as a result of trauma • Sedation for manipulation of fractured limbs or painful/difficult extrication • Rapid Sequence Intubation • Acute psychosis (combative patient) and or excite delirium
Administration and Dosages	**Adult:** • Analgesia 0,3-0,1 mg/kg IV PRN • Sedation/RSI: initial dose 1 mg/kg IV, followed by 0,5mg/kg IV PRN. Initial dose 4mg /Kg IM, followed by 2mg/ kg after 10mins if first dose unsuccessful • Acute psychosis / combative patient: 6-4mg/kg IM or 1mg/kg IV **Paediatrics:** • as per broselow tape
Actions	Ketamine blocks signals from the PNS from reaching the CNS. This results in sedation, analgesia and anaesthesia. IV ONSET: 2–1 minutes , IM ONSET: 10–5 minutes
Contraindication	• Hypersensitivity to the medication • Myocardial ischaemia • Active psychosis or history of schizophrenia • Age < 12 months • Glaucoma or acute globe injury
Caution	• Haemorrhagic stroke • Significant hypertension (> 180 mmHg) • CHF • Thyroid medications
Side effects	**Serious:** • Laryngospasm • emergence • recovery reactions • cardiac ischemia **Less serious:** • Nausea and vomiting • hypersalivation • dysphoria • dystonic type movements • nystagmus • transient hypertension
Additional information	• Monitor for EMERGENCE REACTION from procedural sedation. Complete an incident report and submit to Supervisor for all emergence reactions. • Ketamine is not recommended as first line treatment analgesia in children • Pay close attention to respiratory rate, effort and depth, SpO2, ETCO2 and BP • Lost or stolen Ketamine should be reported immediately to pharmacist and/or duty manager • Give as a slow IV push bolus over 45-60 seconds (Rapid bolus increases risk for apnoea and laryngospasm) • Narcotics and controlled medications policy applicable

Medication	MAGNESIUM SULPHATE
Authorisation	AP EP
Classification	Antiarrhythmic
Presentation	• Ampoule 5 g in 10 ml (1 g per 2 ml) or 2g in 10 ml (1 g per 5 ml)
Indications	• Torsades de pointes (polymorphic ventricular tachycardia) • Life threatening asthma • Seizure associated with eclampsia
Administration and Dosages	**Adult:** • Torsades de pointes: 2-1 g IV bolus over 2-1mins • Eclampsia: 4 g IV over 10mins or 8 g IM (4g in each buttock) • Life threatening asthma: 2-1g IV bolus over 2-1mins **Paediatric:** as per broselow tape
Actions	Acts as a physiological calcium channel blocker and blocks neuromuscular transmission.
Contraindication	• None in cardiac arrest • Known severe adverse reaction
Caution	• Haemorrhagic stroke • Significant hypertension (> 180 mmHg) • CHF • Thyroid medications
Side effects	• Decreased deep tendon reflexes • Respiratory depression • Bradycardia • Hypothermia • Nausea • Vomiting • Thirst • Flushing of skin
Additional information	• Doses are in GRAMS • Monitor blood pressure, respiratory rate, urinary output and for signs of overdose (loss of patellar reflexes, weakness, nausea, sensation of warmth, flushing, drowsiness, double vision, and slurred speech) • Not known to be harmful for short-term intravenous administration in eclampsia, but excessive doses in third trimester can cause neonatal respiratory depression

Medication	LABETALOL
Authorisation	EP
Classification	Beta Blocker
Presentation	Ampoule containing 100mg/20ml
Indications	• Hypertension in pregnancy (Preeclampsia /eclampsia. • Hypertension with Angina and Myocardial infarction. • Hypertension Crisis.
Administration and Dosages	**Adult:** • 20mg IV slowly over 2-1 min. until target pressure (MAX dose 300mg) • Infusion dosing: 2-0,5mg /min
Actions	Nonselective beta blocker with intrinsic sympathomimetic activity; also, alpha blocker Onset of Action:2-5min (IV) Duration of Action: 2-4hr (IV)
Contraindication	• Asthma or obstructive airway disease • Severe bradycardia • Second-degree or third-Degree heart block (without pacemaker) • Cardiogenic shock • Hypersensitivity • Conditions associated with prolonged and severe hypotension • Sinus bradycardia
Caution	None
Side effects	• Postural Hypotension • Headache • Epigastric pain • Nausea • Vomiting • Scalp Tingling
Additional information	None

Medication	METHOXYFLURANE (PENTHROX)
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Inhaled analgesia
Presentation	A hand-held inhaler device for the self-administration of methoxyflurane vapour 3 ml
Indications	• Moderate pain, especially MSK pain
Administration and Dosages	**Adult:** • Have patients inhale until pain is relieved, or until they can't hold the 'whistle' administration device on their own • Patients should not receive more than 6 ml in any 24 hours. Be sure to inquire if patient has been treated with methoxyflurane at all within the last 24 hours • Maximum dose 15 ml/week **Paediatric:** • Children too young to hold the administrator by themselves are too young to receive the medication
Actions	• An extremely potent and highly lipid soluble anaesthetic agent with analgesic effects • ONSET: 6–10 breaths DURATION: constant whilst in use
Contraindication	• Renal impairment and failure • Severe liver disease • Hypersensitivity to fluorinated anaesthetics • Cardiovascular instability: HR: 140 / min RR: 36 / min SPO2: 200 mm Hg • Respiratory depression • Head injury or loss of consciousness • A history of side effects in either patient or relatives • Malignant hyperthermia: patients with known or genetically susceptible to malignant hyperthermia
Caution	• None relevant to a single dose. • Avoid intramuscular administration if patient likely to require thrombolysis
Side effects	• Drowsiness • Hypotension • Bradycardia
Additional information	• 3 ml will last approximately 10–20 minutes. You can tell when the drug has been used as the smell will be gone. • Patients may need to be coached to breathe IN through the mouth, and OUT through the nose in order to receive the medication effectively. After 1–2 breaths the therapeutic effect will wear off. • DO NOT hold the administrator to a patient's mouth if they are unable to do so themselves • Ensure adequate ventilation in clinical environment to avoid passive inhalation • Do not expose medication to temperatures over 40°C especially when used in conjunction with oxygen

Medication	METOCLOPRAMIDE
Authorisation	P AP EP
Classification	Anti-emetic
Presentation	A hand-held inhaler device for the self-administration of methoxyflurane vapour 3 ml
Indications	• Symptomatic treatment of nausea/vomiting • Prevention of nausea/vomiting in patients receiving morphine
Administration and Dosages	**Adult:** • 10mg IV in 100ml NS within 15-10mins , 10mg IM **Paediatric:** • 0,1mg/kg IV or IM
Actions	• An effective anti-emetic which acts centrally by blocking dopaminergic receptors • Enhances GI motility without stimulating gastric secretions • Central dopamine antagonist
Contraindication	• < 1 years old • Renal failure • Pheochromocytoma • GI obstruction / perforation • Recent GI surgery • Allergy to class or drug
Caution	• Haemorrhagic stroke • Significant hypertension (> 180 mmHg) • CHF
Side effects	• Extrapyramidal effects – more common in children and young adults • Cardiac conduction abnormalities following intravenous administration (rare) • Diarrhoea • Rash • Hypotension • Dyspnoea • Anxiety, confusion, and restlessness • Drowsiness & dizziness • Dizziness • Tremors
Additional information	• Should always be administered in separate syringe from morphine – must not be mixed • Metoclopramide should not be used in those patients with hypersensitivity to the drug or its components. Since metoclopramide is structurally related to procainamide, metoclopramide should be used cautiously in patients with a known procainamide hypersensitivity. • Metoclopramide can increase the rate or extent of absorption of other drugs because of accelerated gastric emptying, which increases the contact time with the small bowel where these drugs are absorbed.

Medication	MIDAZOLAM
Authorisation	EP
Classification	Benzodiazepine
Presentation	• Ampoule containing 5 mg in 1 ml
Indications	• Procedural sedation • Sedation of agitated patients • Seizures/status epilepticus • Ketamine emergence and dysphoria (> 5 years only)
Administration and Dosages	**Adult:** • Seizures, Agitated patients: • -0,05 mg/kg IV (5mg average adult) ONCE • Repeat doses only in status epilepticus, maximum cumulative dose 10 mg. • -0,1 mg/kg IM, repeat 0,05 mg/kg ONCE • Repeat doses only in status epilepticus, maximum cumulative dose 15 mg. **Paediatrics:** • Seizures 0,05 mg/kg IV, repeat every 5–3 minutes, maximum cumulative dose 5 mg. • Emergence: 0,025 mg/kg IV, repeat ONCE only after 5–3 minutes • IM: Seizures, Emergence: 0,2 mg/kg IM, repeat dose 0,1 mg/kg ONCE only after 15–10 minutes
Actions	Short acting CNS depressant IV ONSET: 1–5 minutes, PEAK: 2–5 minutes, DURATION: 20–30 minutes IM ONSET: 10–15 minutes, PEAK: 0.5–1 hours, DURATION: up to 6 hrs (mean 2 hrs)
Contraindication	• Hypersensitivity to benzodiazepines • Shock, depressed vital signs or alcohol related altered level of consciousness • Known severe adverse reaction • Respiratory depression
Caution	• Reduce dosage in the elderly, renal or hepatic function impairment or very ill patients • Symptomatic CHF (due to reduced cardiac output), COPD • Myasthenia gravis, Multiple sclerosis • Burns with large surface area
Side effects	• Increase or decrease in BP/HR/RR • Apnoea • Headache • Drowsiness • Dizziness • Hiccups • Erythema • Rash • Pruritis • Hives • Muscle stiffness
Additional information	• Avoid extravasation or intra-arterial injection • Ensure oxygen and resuscitation equipment are available prior to administration • Consider procedural sedation when transcutaneous pacing or synchronised cardioversion. • Lost or stolen Midazolam should be reported immediately to pharmacist in-charge/medical director and duty shift in-charge. • Narcotics and controlled medications policy applicable

Medication	MORPHINE SULFATE
Authorisation	EP
Classification	Opioid Analgesia
Presentation	Ampoule containing 10 mg in 1 ml
Indications	• Chest pain (myocardial ischemia) unrelieved by Nitro-glycerine • Traumatic injury • Burn
Administration and Dosages	**Adult:** • IV/IO/IM: 4-2mg (MAX 10mg). **Paediatric:** • Traumatic pain management <1 YRS • IV/IO : 0,05mg/kg >1YRS • IV/IO:0,1mg/kg (MAX 4mg).
Actions	Narcotic agonist-analgesic of opiate receptors; inhibits ascending pain pathways, thus altering response to pain; produces analgesia, respiratory depression, and sedation; suppresses cough by acting centrally in medulla.
Contraindication	• Hypersensitivity • Paralytic ileus, • Toxin-mediated diarrhoea, • Respiratory depression • Acute or Severe bronchial asthma. • Upper airway obstruction • Head injuries • Brain tumours • Deliriums tremens • Seizure disorders • Cardiac Arrhythmia
Caution	• Use with caution in the elderly/young • Renal or hepatic impairment or pregnancy – use smaller doses carefully and titrate to effect • Use with GREAT CAUTION in chest injuries, especially with any respiratory difficulty (although if respiration is inhibited by pain, analgesia may improve respiratory status). • Morphine frequently induces nausea or vomiting which may be potentiated by the movement of the ambulance. Titrating to the lowest dose and administering slowly to achieve analgesia will reduce the risk of vomiting. The use of an anti-emetic should also be considered.
Side effects	• Respiratory depression • Decreased blood pressure • Drowsiness • Pupillary constriction • Histamine release/rash • Increased intracranial pressure
Additional information	• Unused morphine must be discarded of safely into a clinical waste bin • Lost, stolen or missing morphine must be reported immediately to a pharmacist in-charge /medical director / duty shi in-charger. • Most patients do not require doses greater than 10 mg/hr, but terminal patients may require > 100 mg/hr

Medication	NALOXONE
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Narcotic antagonist, antidote
Presentation	• Ampoule containing naloxone hydrochloride 400 mcg in 1 ml • Nasal spray naloxone hydrochloride 4mg in nasal single dose
Indications	• Suspected opioid overdose producing respiratory • Cardiovascular • Central nervous system depression
Administration and Dosages	**Adult:** • IV/IM: 400 mcg (0,4 mg) repeat after 3 min prn to a Max 2 mg • IN (Intra Nasal): 4 mg repeat after 3 min prn to a Max 20 mg **Paediatrics:** • V/IM: 10 mcg/kg repeat after 3 min prn to a Max 2 mg.
Actions	Competitive narcotic receptor antagonist IV ONSET: 2 – 10 minutes
Contraindication	Known severe adverse reaction
Caution	None
Side effects	• In patients who are physically dependent on opiates, naloxone may precipitate violent withdrawal symptoms, including cardiac dysrhythmias. Ensure regular re-assessment of ventilation and circulation. • Seizures
Additional information	IV administration is preferred. If access is impossible, naloxone may be administered intramuscularly, undiluted (into the outer aspect of the thigh or upper arm), but absorption may be unpredictable. • When used, naloxone s effects are short lived – respiratory and cardiovascular depression can recur with fatal consequences • All cases of opioid overdose should be transported to hospital, even if the initial response to naloxone has been good. For intranasal spray: • Before administration, ensure insert device nozzle in either nostril, and provide support to back of neck to allow head to tilt back. • Use each nasal spray only 1 time; do NOT test or prime before use; press firmly on device plunger to administer dose and remove it from nostril after use. • Do not attempt to reuse naloxone intranasal; each spray contains a single dose of naloxone and cannot be reused. • Readminister using a new nasal spray every 2-3 minutes if the patient does not respond or responds and then relapses into respiratory depression • Administer in alternate nostrils with each dose

Medication	Oxygen
Class	Gas.
Descriptions	Odourless / Tasteless / Colourless gas necessary for life.
Presentation	Medical gas: D, E or F cylinders, coloured black with white shoulders. CD cylinder: White cylinder.
Administration	Inhalation via: High concentration reservoir (non-rebreather) mask / Simple face mask / Venturi mask / Tracheostomy mask / Nasal cannulae / CPAP device / Bag Valve Mask. (CPG: Oxygen is used extensively throughout the CPGs).
Indications	Absent / Inadequate ventilation following an acute medical or traumatic event. SpO_2 < 94% adults and < 94% paediatrics. SpO_2 < 92% for patients with acute exacerbation of COPD. SpO_2 < 90% for patients with acute onset of Pulmonary Oedema.
Contra-Indications	Bleomycin lung injury.
Usual Dosages	Adult: Cardiac and respiratory arrest or sickle cell crisis; 100%. Life threats identified during primary survey; 100% until a reliable SpO2 measurement obtained then titrate O2 to achieve SpO2 of 94% - 98%. For patients with acute exacerbation of COPD, administer O2 titrate to achieve SpO2 92% or as specified on COPD Oxygen Alert Card. All other acute medical and trauma titrate O2 to achieve SpO2 94% - 98%. Paediatric: Cardiac and respiratory arrest or sickle cell crisis; 100%. Life threats identified during primary survey; 100% until a reliable SpO2 measurement obtained then titrate O2 to achieve SpO2 of 96% - 98%. Neonatal resuscitation (< 4 weeks) consider supplemental O2 (≤ 30%). All other acute medical and trauma titrate O2 to achieve SpO2 of 96% - 98%.
Pharmacology / Action	Oxygenation of tissue/organs.
Side effects	Prolonged use of O2 with chronic COPD patients may lead to reduction in ventilation stimulus.
Additional information	A written record must be made of what oxygen therapy is given to every patient. Documentation recording oximetry measurements should state whether the patient is breathing air or a specified dose of supplemental Oxygen. Consider humidifier if oxygen therapy for paediatric patients is > 30 minutes duration. Caution with paraquat poisoning, administer Oxygen if SpO_2 < 92%. Avoid naked flames, powerful oxidising agent.

Medication	PARACETAMOL
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Analgesic and antipyretic
Presentation	• Tablets 500 mg • Vial containing paracetamol 1 g in 100 ml • Suppository 125mg, 250mg
Indications	• Relief of minor or moderate pain • High temperature
Administration and Dosages	**Adult:** • PO: 1 g every 4 – 6 hours up to a maximum of 4 g in 24 hrs • IV: 1 g slow infusion over 15 minutes. Max. of 4 g in 24 hrs **Paediatrics:** • PR > 5 YRS 250mg suppository every 4 – 6 hours up to a maximum of 1 g in 24 hrs • PR <5 YRS 125mg suppository every 4 – 6 hours up to a maximum of 750 mg in 24 hrs • IV paracetamol 10mg /kg, not recommended below 33kg
Actions	• Antipyretic, Analgesic
Contraindication	• Known paracetamol allergy • Do NOT give further paracetamol if a product containing paracetamol has already been given within the last four hours or if the maximum cumulative daily dose has been given already
Caution	• None relevant to a single dose. • Avoid intramuscular administration if patient likely to require thrombolysis
Side effects	• Occasionally IV paracetamol may cause systemic hypotension if administered too rapidly. • Should be given over 5 – 10 minutes.
Additional information	• Pediatrics patients with fever, not responding to oral paracetamol shall be transported to hospital for further evaluation. Any IV paracetamol that remains within the giving set can be flushed using %0,9 saline. • Take care to ensure that air does not become entrained into the giving set; if there is air in the giving set ensure that it does not run into the patient with further fluids.

Medication	PETHIDINE HYDROCHLORIDE
Authorisation	EP
Classification	Opioid Analgesia
Presentation	Ampoule : 50mg /2 mL - 100mg/2mL
Indications	• Used for the treatment of moderate to severe pain.
Administration and Dosages	**Adult:** • IM/SC: 25mg -100mg. Max dosage is 400mg in 24hrs. • IV slow: 25mg -50mg. Max dosage is 200mg in 24hrs. **Paediatric** • IM: 0.5mg/Kg - 2mg/Kg every 4 hours.
Actions	Works by binding to opioid receptors in the CNS reducing the transmission of pain signals.
Contraindication	• Hypersensitivity • Head Injury. • Respiratory Depression. • Convulsive Disorders • Coma. • Patient on monoamine oxidase inhibitors therapy • Hypertension • Myocardial Ischemia. • Acute Asthma or Difficulty in Breathing • Renal Impairment.
Caution	• Pethidine carries the risk of dependency, tolerance and potential for abuse. • Make sure to take the patient history of substance abuse or addition.
Side effects	• Drowsiness. /Dizziness • Respiratory depression. • Hypothermia • Convulsion in case of overdose • Bradycardia, Tachycardia or palpitations. • Headache, confusion or hallucinations.
Additional information	• Unused pethidine must be discarded of safely into a clinical waste bin • Lost, stolen or missing pethidine must be reported immediately to a pharmacist in-charge /medical director / duty shift in-charger. • Most patients do not require doses greater than 10 mg/hr, but terminal patients may require > 100 mg/hr

Medication	RACEMIC EPINEPHRINE 2.25%
Authorisation	AP EP
Classification	Sympathomimetic Bronchodilators
Presentation	• Vial 0.5mL containing 2.25%
Indications	• Used for temporary relief of symptoms associated with bronchial asthma and croup in children.
Administration and Dosages	**Adult:** • 0.5mL solution diluted in 3 mL normal saline over 15 minutes. ONCE **Paediatrics > 4 yrs:** • 0.5mL solution diluted in 3 mL normal saline over 15 minutes. ONCE **Paediatrics < 4 yrs:** • 0.05mL/kg solution diluted in 3 mL normal saline over 15 minutes. ONCE
Actions	Causes the smooth muscles of the airway to relax, which helps to open up the airways and improve
Contraindication	They are cautions
Caution	• Cardiac disease • hypertension, • hyroid disease • diabetes • pregnancy & lactating mothers
Side effects	• Rapid Heart Rate • Chest pain (angina). • Tremor. • Ventricular Fibrillation. • Pulmonary oedema • Headache and restlessness.
Additional information	• Don't use this medication if patient received MAO inhibitors in the last 14 days. • Don't use product if its brown or cloudy or pinkish.

Medication	RINGER LACTATE
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Crystalloid solution
Presentation	• 500 ml in a Plastic bottle of Compound Sodium Lactate (Ringers Lactate)
Indications	• Fluid and electrolyte replacement. • Trauma or burn patients. • Hypovolemia.
Administration and Dosages	**Adult: IV/IO:** • 40ml/kg for 3 – 5 minutes to keep vein open. **Paediatric:** • IV/IO: 10 ml/kg for 3 – 5 mins or to keep vein open.
Actions	• Increases vascular fluid volume, raising cardiac output and improves perfusion
Contraindication	• DKA coma and pre-coma • Neonates • Severe Hypertension • Hypervolemia
Caution	• Avoid use in limb crush injury • Renal failure • Liver failure
Side effects	• Electrolytes disturbance. • Heart failure in patients with cardiac disorder. • Hyperhydration in patients with pulmonary oedema.
Additional information	• The volume of compound sodium lactate IV/IO infusion needed is 3 times as great as the volume of blood loss • Sodium lactate has NO oxygen carrying capacity • To calculate the amount should be given to burn patient the formula is 4X patient weight in Kg X % of total body surface area burned. • Administration is given too fast can lead to water and sodium overload with a risk of oedema.

Medication	SALBUTAMOL
Authorisation	EMT · EMT ADVANCE · P · AP · EP
Classification	Sympathetic Agonist
Presentation	• Nebule containing 2.5 mg in 2.5 ml
Indications	• Bronchospasm • Exacerbation of COPD • Respiratory distress following submersion incident
Administration and Dosages	(dilute with 2-4ml sterile water) **Adult:** • Nebulised: 5 mg with 6 – 8 l/min oxygen. Repeat every 5 minutes prn. Maximum cumulative dose 15 mg **Paediatric:** • Nebulised: 5 – 12 yrs: 5 mg with 6 – 8 l/min oxygen. Repeat every 5 minutes prn. Maximum cumulative dose 15mg • Nebulised: < 5 yrs: 2.5 mg nebulised with 6 – 8 l/min oxygen. Repeat every 5 minutes prn. Maximum cumulative dose 15mg
Actions	Salbutamol is a selective β2 adrenoreceptor stimulant drug. It's a Bronchodilator and relaxes the smooth muscle. ONSET: immediate
Contraindication	Known severe adverse reaction
Caution	• Hypertension, Angina, Cardiac tachyarrhythmias, Coronary insufficiency, Overactive thyroid, Late pregnancy (can relax uterus) • Severe hypertension may occur in patients on beta blockers and half doses should be used unless there is profound hypotension
Side effects	• Tachycardia • Tremors • Tachyarrhythmias • High doses may cause Hypokalaemia.
Additional information	• In acute severe or life-threatening asthma ipratropium should be given after the first dose of salbutamol • In more severe attacks the use of steroids by injection or orally and • further nebuliser therapy will be required. Do not be reassured into a false sense of security by an initial improvement after salbutamol • Beware of the "silent chest asthmatic" as severe bronchospasm may present with absent air entry and no evidence of wheezing. • If this occurs and the patient requires assisted ventilations, consider administration of Epipen.

Medication	0.9% SODIUM CHLORIDE
Authorisation	EMT, EMT ADVANCE, P, AP, EP
Classification	Fluid replacement therapy
Presentation	250 ml and 100 ml packs of sodium chloride 0.9%
Indications	• Trauma and Medical conditions with signs of poor perfusion • Flush to confirm and maintain patency of IV/IO cannula & following drug administration • Symptomatic Hyperglycaemia (> 250 mg/dl, > 13.9 mmol/l) (Paramedic and above)
Administration and Dosages	**Adult: > 12 yrs:** Dosage is based on medical condition and response to treatment **Paediatric: < 12 yrs:** Dosage is based on age, weight, medical condition and response to treatment.
Actions	• Increases vascular fluid volume • Raising cardiac output • Improving perfusion
Contraindication	Pulmonary Oedema
Caution	• Fluid replacement in dehydration should occur over hours • In paediatric patients su ering from heart failure, renal failure and diabetic ketoacidosis smaller doses are given ml/kg. • In DKA administer ONCE only over 15 minutes
Side effects	• Over infusion precipitates pulmonary oedema causing breathlessness
Additional information	• Despite a lack of evidence demonstrating any significant beneficial effects, prehospital fluid therapy has become established practice. **DO NOT DELAY TRANSPORT** • Routine use of fluid may be detrimental to patient.

Medication	TRANEXAMIC ACID
Authorisation	P AP EP
Classification	Anti-fibrinolytic
Presentation	• Ampoule containing 500 mg tranexamic acid in 5 ml (100 mg/ml)
Indications	• Severely bleeding trauma patients, SBP< 90mmHg &/or HR >110 beats/min • Epistaxis, in anticoagulant patients
Administration and Dosages	IV/ IO in 100mL saline infused over 10 minutes (initiated within 3 hours of injury) **Adult:** • 1 g - ONCE only **Paediatric (< 12 yrs.):** • 20 mg/kg - ONCE only
Actions	anti-fibrinolytic which reduces breakdown of blood clots
Contraindication	• First trimester of pregnancy. • History of seizures. • renal impairment. • > 3hrs from injury • Known sensitivity to TXA • Previous DVT or PE
Caution	• Head injury
Side effects	• Nausea, vomiting and Diarrhoea. • Convulsions. • Hypotension with Rapid injection • Arterial or Venous Thrombosis
Additional information	• There is good data that this treatment is safe and effective (giving a 9% reduction in the number of deaths in patients in the CRASH2 trial) • Topical use in epistaxis: 500mg – 1g applied to packing

Intervention Medication Matrix

SECTION 7

MEDICATION ADMINISTRATION

CLINICAL LEVEL / MEDICATIONS	EFR	EMT	EMT-A	P	AP	EP
Oxygen						
Biofreeze						
Naloxone IN						
Paracetamol PO						
Dexamethasone PO						
Aspirin PO						
Paracetamol PR/ IV						
Glucose gel buccal						
GTN SL						
Combivent						
Salbutamol						
water for injection						
Penthrox						
Dextrose %10						
Ringer Lactate						
Sodium Chloride						
Epinephrine 1:10,000 PFS						
EpinPen 0,3mg, 0,15mg						
Diphenhydramine IM						
Glucagon SC/IM						
Naloxone IM/IV						
Amiodarone						
Hydrocortisone IM						
Diclofenac IM						
Clopidogrel PO						
Atropine						
Adenosine						
Hyoscine						
Metoclopromide						
Epinephrine 1:1000 amp						
Tranexamic Acid						
Furosemide						
Racemic Epinephrine						
Dopamine						
Labetalol						
Morphine						
Pethidine						
Haloperidol						
Midazolam						
Diazepam IM/IV/PR						
ketamine						

AIRWAY & BREATHING MANAGEMENT

TYPE OF INTERVENTION	EFR	EMT	EMT-A	P	AP	EP
Oxygen therapy						
nebulization						
Open Airway Maneuvers						
Suction						
Airway Adjuncts						
End- tidal CO2 monitoring						
Bag Mask Ventilation						
Largygeal Tube insertion						
CPAP						
Endotracheal intubation						
Invasive Ventilation						
Non-Invasive Ventilation						
Rapid Sequence Intubation						
Foreign Body Removal						

CARDIAC

12 lead ECG						
AED & CPR adult &						
pediatrics						
Defibrillation						
Death Recognition &						
Declaration “Black“						
Death Recognition &						
Declaration “Red“						
Cardioversion						

TRAUMA

Type of intervention	EFR	EMT	EMT-A	P	AP	EP
Triaging - Establishing		■	■	■	■	■
Priorities		■	■	■	■	■
Rapid extrication		■	■	■	■	■
KED		■	■	■	■	■
Spinal stabilization		■	■	■	■	■
C-collar application		■	■	■	■	■
Pelvic Binder		■	■	■	■	■
Scoop stretcher		■	■	■	■	■
Splinting		■	■	■	■	■
traction		■	■	■	■	■
Sling application		■	■	■	■	■
Logroll		■	■	■	■	■
Helmet stabilization/ removal		■	■	■	■	■
Needle decompression					■	■

HEMORRHAGE CONTROL

	EFR	EMT	EMT-A	P	AP	EP
Tourniquet use		■	■	■	■	■
Nasal packing					■	■

OTHERS

	EFR	EMT	EMT-A	P	AP	EP
Childbirth		■	■	■	■	■
intramuscular injection		■	■	■	■	■
intravenous cannulation		■	■	■	■	■
intraoesseous cannulation			■	■	■	■
Procedure sedation						■

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